

30 Colour Tests

B Widdop

Introduction

The bulk of the material in this chapter derives from that contributed by the late HM Stevens to the second edition of this publication, which was modified and expanded by Wayne Jeffrey in the third edition. The main addition in this latest version is an appendix dealing with tests for metals and anions that are not covered by the general scheme.

Colour tests (sometimes referred to as chemical spot tests) provide toxicologists and drug analysts with one of the first tools for the presumptive identification of drugs and poisons. These colour tests are most usefully applied to pharmaceuticals and scene residues, and to a lesser extent to biological fluids such as stomach contents, urine. They are used to place the unknown into a specific class of compounds or to eliminate categories or classes of compounds. These colour tests remain popular for many reasons. They are simple to perform, use minimal reagents, are inexpensive and give results that can be viewed by the naked eye. They appeal particularly in parts of the developing world where laboratory facilities tend to be very limited. In many instances they can also be used as thin-layer chromatography (TLC) location reagents applied by spraying or dipping (see Chapters 1, 11, and 39). This chapter describes the colour tests cited in the monographs. For some substances, the colour reaction with a particular chemical reagent may be quite specific, but it is much more common for the colour to be produced by a class of compounds. Moreover, compounds that do not fall into the class may also give colours. For some of the tests, the colour reactions can be correlated with certain aspects of the chemical structure of a compound or group of compounds. However, anomalous responses often occur that cannot be explained on that basis. Some of these are noted in the colour tests described below, but it should be borne in mind that many others may be found. It follows that colour tests are only an indication of the presence of a compound or class of compounds and that all tests must be confirmed by more specific methods. This is especially important in forensic cases!

The colour tests included here range from those that rely on reactions with certain functional groups (e.g. Folin–Ciocalteu for phenols), those that are almost specific for a given group (e.g. FPN reagent for phenothiazines) through to those that give diagnostic colours with a wide range of compounds (e.g. Mandelin's test and the Marquis test).

Interpretation of colour tests

Colour descriptions

Colours exhibited by these tests cannot be described with any accuracy. They may vary in intensity or tincture with the concentration of compounds in the test samples and the presence of extraneous material. In addition, their assessment is always a subjective one, even in people with normal colour vision. Some of the complexes formed are unstable such that the colour changes or fades with time.

Effects of ionic form

Salts may give colours different from those of the corresponding acid or base. In general, free acids or bases that have been isolated from the test material by an extraction process give better colours than their salts. The colour of a salt may be modified by the nature of the other ions present. For example, all hydrochloride salts give a red colour in Mandelin's test

and a blue colour with Koppanyi–Zwicker reagent (prior to adding pyrrolidine). Basic salts of weak acids may produce different colours because of a change in pH. Where a compound has been extracted from biological material, these factors should not create any difficulty, since it will be present in the form of the base. However, when applying the tests to pharmaceutical preparations, where the compounds are usually present as salts, this can cause problems. To overcome this, the material can be extracted in much the same way as for biological samples to derive the free base. Bromide and iodide salts can be converted into the nitrate before testing, which gives the same colour as the base, by the following method:

- To 0.5 mL of a 1% (w/v) solution of the salt in dilute acetic acid, add one drop of an 8% (w/v) solution of silver nitrate followed by one drop of a 2% (w/v) solution of sodium chloride to remove excess silver.
- Centrifuge to separate the precipitated silver halide and use the supernatant liquid, either as a solution or evaporated to dryness where necessary, for the colour tests.

The colours that are recorded in the tables and monographs are usually those obtained by testing either the free acid or the free base.

Use of the colour tests lists

The system adopted uses ten basic colours: the spectral colours (red, orange, yellow, green, blue and violet), together with pink, brown, grey and black. Where there is a variation in hue, this is indicated by combining two colours (e.g. red–brown). The second-named colour is considered to be the dominant one and is the main colour used in the lists. For example, red–brown is listed under brown, whereas brown–red is listed under red. When interpreting results, it is often necessary to search the lists given under two main colours (e.g. for red–brown, the lists under both red and brown should be consulted). This takes account of the subjective nature of colour assessment. An arrow between two colours (e.g. red→brown) indicates that the colour changes during the course of the test. In the monographs, the notation brown/red is used where there are two parts to a test that produce two colours. Occasionally, the colour displayed by a test solution in reflected light may be different from that in transmitted light, in which case the solution is described as dichroic. A combined colour may be obtained when more than one drug is present or the drug itself is coloured, which limits the value of the tests for biological samples.

Practical points

Performing the colour tests

The tests are carried out either in clear glass test-tubes or on white glazed porcelain tiles (spotting tiles), which give a uniform background against which the colours can be assessed. For drugs, the tests are designed to work on about 1 mg, either as the solid form or as a dried extract of this amount (see below), unless stated otherwise. Solutions should be made in water unless otherwise stated. Where an instruction, time, temperature, etc., appears in brackets after the drug name, such as (add water), (15 s) or (slowly at 100°C), this indicates a change in the test procedure for that particular drug.

The following recommendations are most important:

- A sample known not to contain the compound of interest should be tested at the same time as the test sample. This enables a comparison of the colours produced by the sample and by the reagent blank. Ideally, the blank sample should have the same matrix as the test sample (e.g. for urine tests use analyte-free urine), since this takes account of the effects of extraneous materials. Otherwise, water is usually adequate.
- Before making a final decision on the result of a test, the reaction of the unknown should be compared with that of a reference substance tested under exactly the same conditions.

Validation of a colour test

It is essential to validate all tests and test reagents for sensitivity and specificity; O'Neal *et al.* (2000) have outlined a suitable method for a chemical spot test.

Application of colour tests to sample extracts

Several solvent extraction schemes have been devised to fractionate compounds on the basis of their acidic, neutral or basic characteristics (see Chapters 1 and 10). The tests listed in Table 30.1 can be applied to the evaporated extracts.

Table 30.1 Tests that can be applied to the evaporated extracts	
Fraction	Test
Strong acid	Aromaticity
	Ferric chloride
	Folin-Ciocaltaeu reagent
	Liebermann's reagent
	Millon's reagent
	Nessler's reagent
Weak acid	Aromaticity
	Coniferyl alcohol
	Diazotisation
	Ferric chloride
	Folin-Ciocaltaeu reagent
	Koppanyi-Zwicker reagent
	Liebermann's reagent
	Mercurous nitrate
	Millon's reagent
Neutral	Nessler's reagent
	Aromaticity
	Furfuraldehyde
	Koppanyi-Zwicker reagent
	Liebermann's reagent
	Mercurous nitrate
Basic	Nessler's reagent
	Amalic acid test
	<i>p</i> -Dimethylaminobenzaldehyde
	Ferric chloride
	Formaldehyde-sulfuric acid
	Forrest reagent
	FPN reagent
	Liebermann's reagent
	Mandelin's reagent
	Marquis reagent
	Nessler's reagent
	Sulfuric acid

Table 30.2 is formulated to give a quick lead to those tests that can be applied to detect some of the most important drug groups and other poisons.

Table 30.2 Indication of which tests can be applied to detect some of the most important drug groups and other poisons	
Substance/functional group	Useful tests
Alcohols	Potassium dichromate
Alkaloids and nitrogenous bases	Dragendorff's reagent
Amides (aliphatic)	Nessler's reagent
Aldehydes (aliphatic)	Schiff's reagent
Amfetamines	See Appendix 30.1
	Sodium nitroprusside-acetone
Antidepressants	Marquis test
Barbiturates	Dille-Koppanyi reagent
	Koppanyi-Zwicker reagent
	Mercurous nitrate
	Vanillin reagent
	Zwicker reagent
Benzodiazepines	Formaldehyde-sulfuric acid
Cannabis	Duquenois reagent
Carbamates (non-aromatic)	Furfuraldehyde
Cocaine	Cobalt thiocyanate
	<i>p</i> -Dimethylaminobenzaldehyde
	Mandelin's test
	Scott's test
Chlorinated phenols	Nitric acid (fuming)
Chlorinated hydrocarbon insecticides	Nitric-sulfuric acid
Cyanide	Ferrous sulfate (B)
	Sodium picrate
Cyanide groups	Sodium picrate
Dithiocarbamates	Sodium nitroprusside
Ergot alkaloids	<i>p</i> -Dimethylaminobenzaldehyde
Halogenated hydrocarbons	Fujiwara test
Imides	Koppanyi-Zwicker test
Ketones	Sodium nitroprusside
Methadone	Cobalt thiocyanate
	Mandelin's test
	Marquis test
	Tetrabromophenolphthalein ethyl ester
Mono-substituted pyridine ring	Cyanogen bromide
Nitrates and nitrites	Ferrous sulfate
Opiates	See Appendix 30.2
Oxidising agents	Diphenylamine
Paraquat/diquat	Sodium dithionate
Phencyclidine	Cobalt thiocyanate
	<i>p</i> -Dimethylaminobenzaldehyde
	Tetrabromophenolphthalein ethyl ester
Phenols	<i>p</i> -Dimethylaminobenzaldehyde
	Ferric chloride
	Folin-Ciocaltaeu reagent
	Millon's reagent

Table 30.2 continued

Substance/functional group	Useful tests
Phenothiazines	Ferric chloride
	Formaldehyde-sulfuric acid
	Forrest reagent
	FPN reagent
Phenylpyrazolines	Nitrous acid
Primary aromatic amines	Coniferyl alcohol
	Diazotisation
Primary and secondary amines	Dragendorff's reagent
	Simon's test
Propoxyphene	Cobalt thiocyanate
	Froehde's reagent
	Liebermann's test
	Tetrabromophenolphthalein ethyl ester
Quaternary ammonium compounds	Tetrabromophenolphthalein ethyl ester
Quaternary amines	Dragendorff's reagent
Quinines	Cobalt thiocyanate
	Thalleioquin
Quinones	Methanolic potassium hydroxide
Reducing agents	Benedict's reagent
Salicylates	Ferric chloride
	Trinder's reagent
	Antimony pentachloride
Steroids	Naphthol sulfuric acid
	Sulfuric acid
	Copper sulfate
	Koppanyi-Zwicker reagent
Sulfonamides	Mercurous nitrate
	Nitrous acid
	Palladium chloride
	Sodium nitroprusside
Sulfur containing	Dragendorff's reagent
	Tetrabromophenolphthalein ethyl ester
Tertiary amines	

Additional information

Colour reactions given by narcotics and amfetamines with four of the so-called 'alkaloid colour reagents' (Marquis, Mecke's, Froehde's and Mandelin's) are listed in Appendix 30.1 and Appendix 30.2. Three colour tests have been developed for gamma-hydroxybutyric acid (GHB) and its precursor gamma-butyrolactone (GBL; see Appendix 30.3).

Colour test methods

Caution: the following lists of colour tests and drugs tested are not exhaustive; the omission of a compound from a list does not indicate that no response is given, but that it may not have been tested.

Amalic acid test (test for xanthines)

Method

Add to the sample a few drops of 10 mol/L hydrochloric acid followed by a few crystals of potassium chlorate, and evaporate the mixture to dryness. Observe the colour of the residue then add 2 or 3 drops of 2 mol/L ammonium hydroxide and again observe the colour.

Table 30.3 Colours with amalic acid

Colour	Compound
Red (→violet)	Bufylline, caffeine
Pink (→violet)	Pentoxifylline
Orange (→violet)	Acepifylline piperazine, bamifylline, xanthinol nicotinate
Pink-orange (→violet)	Fenetylline, pentifylline
Yellow (→pink)	Etamiphylline
Yellow (→violet)	Diprophylline, proxyphylline, theobromine, theophylline

Indications

A red, pink, orange or yellow residue, which changes to pink, red or violet after the addition of ammonium hydroxide, indicates the presence of a xanthine (Table 30.3).

Ammoniacal silver nitrate

Reagent

To 20 mL of 0.1 mol/L silver nitrate add sufficient strong ammonia solution to dissolve the initial precipitate.

Method

Dissolve the sample in a minimum amount of water, with the addition of ethanol if necessary, add an equal volume of the reagent and note any colour that develops. Heat the mixture in a water-bath at 100°C for 30 s.

Indications

Red, yellow, brown or black colours (especially at room temperature) indicate potent reducing power, which occurs when adjacent carbon atoms in a ring each bear a hydroxyl group (Table 30.4). There is no response when the hydroxyl groups are *meta* to each other, but there is some restoration of reducing power when they are *para* to each other. Some colour production is also obtained with ethynyl bonds, but not with ethylenic bonds. Ethchlorvynol and ethinylestradiol both give a white precipitate that turns yellow on heating. Carbidopa gives a silver mirror on heating.

Table 30.4 Colours with ammoniacal silver nitrate

Colour at room temperature	Compound	Colour at 100°C
Red	Isoetharine	Brown-orange
	Hexoprenaline (→brown→black)	
	Isoprenaline (→red-brown)	Brown
	Rimiterol	Brown
Yellow	Ethinamate	Brown
	Levodopa (→brown)	Black
Grey-yellow	Hydroquinone	Brown
Brown		
Red-brown	Adrenaline	
	Methyldopa	Black
Orange-brown	Dopamine	Black
	Methyldopate	Orange-brown
Grey		
Red-grey	Protokylol	Brown
Black	Ascorbic acid	
	Benserazide	–
	Dobutamine	–
	Dodecyl gallate	–
	Noradrenaline	–

Antimony pentachloride

Reagent

Dry some antimony trichloride over phosphorus pentoxide, melt the dried material (m.p. 73°C), and pass dry chlorine gas into the melt until a yellow fuming liquid is obtained. Add this liquid to about 10 times its volume of chloroform, filter the solution into a dark glass-stoppered bottle and store in a desiccator.

Method

Place a drop of an ethanolic solution of the sample on a filter paper, add a drop of the reagent and dry in a current of warm air. Alternatively, the test may be carried out by adding a drop of the reagent to the sample on a white tile.

Indications

Various colours are obtained with the cardiac glycosides, their aglycones and certain oestrogens and corticosteroids (Table 30.5). No colour is obtained with beclometasone, cortisone, fluocinolone, fludroxycortide, prednisolone, prednisone, progesterone, testosterone or triamcinolone.

Aromaticity

Method 1

Place a portion of the sample in each of two ignition tubes, and to one tube add some solid sodium hydroxide. Heat both tubes carefully, allow the water vapour to escape, insert into the vapours in each tube an open capillary tube that contains Marquis reagent, and observe the colour of the reagent.

Indications

Red or orange colours indicate that the sample is aromatic in nature. The colours probably result from the liberation of traces of aromatic hydrocarbons, phenols, etc. Colours obtained after heating with sodium hydroxide generally indicate the presence of aromatic acids. Colours obtained after heating without sodium hydroxide generally indicate the presence of phenols, phenolic acids and aldehydes that contain more than one hydroxyl group.

A negative result does not necessarily imply that the substance is non-aromatic.

Method 2

Add 2 or 3 drops of concentrated nitric acid to the sample, heat in a water-bath at 100°C for 1 min, cool the mixture, dilute 3–4 times with water and make the solution alkaline by the addition of a 40% (w/v) solution of sodium hydroxide.

Table 30.5 Colours with antimony pentachloride

Colour	Compound
Red	Dienestrol, diethylstilbestrol
Orange	Cholesterol (→brown), desoxycortone, dydrogesterone, fludrocortisone, hydrocortisone, hydroxyprogesterone, strophanthin-K (→red)
Yellow	Alfadolone
(→brown)	Androsterone, digitoxigenin, digoxigenin
(→brown→black-violet)	Digitoxin, digoxin, lanatoside C, ouabain (very weak)
Green-yellow	Fluocortolone
Green	Betamethasone (→brown), dexamethasone, mestranol, pancuronium
Brown	Carbenoxolone, dimethisterone, estradiol, estriol, estrone, ethinylestradiol (→black), fluoxymesterone, norethandrolone, norethisterone, oxymetholone, rotenone
Orange-brown	Enoxolone (→violet)
Green-brown	Noretynodrel

Table 30.6 Colours with aromaticity test, method 2, obtained on addition of alkali to acid solution

Colour of acid solution	Colour after addition of alkali	Compound
Colourless	Red	Clobazam (heat for 3 min with acid)
	Orange	Butanilcaine, tolazoline
	Yellow	Amprolium, atropine methobromide, hyoscine butylbromide, hyoscine methobromide, ketoprofen, pipazethate, tetrahydrozoline, tetramisole, trimetaphan
	Violet	Atropine methonitrate (transient), hyoscine methonitrate (transient)
	Brown	Isopropamide
Yellow	Red	Aminacrine, benzonatate, tetracaine (amethocaine), trimethoprim
	Orange-red	Dextromethorphan, haloperidol
	Orange	Amicarbalide, carbocromen, dyclonine, glibenclamide, levallorphan, metocurine, padimate, propanidid, quinuronium, salinazid
	Brown	Dibromopropamide, dichlorophen tubocurarine
Red	Brown-violet	Dequalinium

Indications

A change from colourless or yellow in acid solution to darker colours (e.g. orange or red-orange) after the addition of sodium hydroxide indicates the presence of a benzene ring in the molecule, probably though the production of a nitrophenol or other nitro compound.

Certain compounds (e.g. diazepam, methaqualone) give a negative result. Orange colours are given by certain non-aromatic corticosteroids (e.g. cortisone), by substances that contain sulfur and by compounds that already contain an aromatic nitro group (e.g. nifursol).

Colour changes are given in Table 30.6.

Certain substances give distinct colours with cold nitric acid, but the colours fade on heating; these are listed in Table 30.7.

Benedict's reagent

Reagent

Dissolve 1.73 g of copper sulfate in 10 mL of water. Dissolve 17.3 g of trisodium citrate and 10 g of anhydrous sodium carbonate in 80 mL of water with the aid of heat; pour this solution into the copper sulfate solution and dilute the mixture to 100 mL.

Method

Add 0.5 mL of the reagent to the sample and heat in a water-bath at 100°C for 3 min.

Indications

The formation of red cuprous oxide occurs with strong reducing agents, such as ascorbic acid, dithionites, certain phenolic compounds that contain two hydroxyl groups *para* to each other, and compounds that contain at least four hydroxyl groups on a non-aromatic ring (e.g. glucose, tetracyclines).

Table 30.7 Colours with aromaticity test, method 2, obtained on addition of cold nitric acid to sample, which fade on heating

Colour	Compound
Red	Aminacrine (15 s), clozapine, dropropizine, medazepam, trimethoprim
Brown	Metocurine
Pink-brown	Diethylthiambutene (changing to green)
Black	Tubocurarine

A weak response (orange–brown or brown colours) is given by streptomycin, hydroxylamine and substituted hydrazines (e.g. phenelzine). No colour is obtained with beclometasone, cardiac glycosides and estriol (two hydroxyl groups) or clindamycin (three hydroxyl groups).

Carbon disulfide

Method

Mix the sample with 1 mL of water and 0.1 mL of a 1% (w/v) solution of sodium tetraborate, add 0.2 mL of a 10% v/v solution of carbon disulfide in ethanol and heat in a water-bath at 100°C for 3 min; cool the solution and add 3 drops of 0.1 mol/L silver nitrate.

Indications

A brown colour indicates the presence of a dithiocarbamate, which suggests that the original substance was an aliphatic or heterocyclic primary or secondary amine. The original sample should be tested to ensure that it does not give a brown colour with silver nitrate alone.

Chromotropic acid

Reagent

1. Dissolve 20 mg of chromotropic acid in 10 mL of concentrated sulfuric acid.
2. Dissolve 1 g of sodium nitrite in 10 mL of concentrated sulfuric acid.

Method 1

Add a small amount of sample, either solid or in solution, to 1 mL chromotropic acid reagent. Note any colour that may be produced, and then add the solution dropwise to 0.5 mL of water, with cooling. Substances that give a colour with cold sulfuric acid must be excluded.

Indications

See Table 30.8.

Table 30.8 Colours with chromotropic acid

Colour	Compound
Red (before dilution)	Formaldehyde, paraformaldehyde (reacts slowly)
Violet (after dilution)	Hydrochlorothiazide, hydroflumethiazide

Method 2 (for chlorophenoxy herbicides)

Add 1 mL of 1 mol/L hydrochloric acid to 10 mL of sample and extract with 20 mL of toluene for 5 min. Centrifuge for 5 min and remove the toluene layer. Repeat with a further 20 mL of toluene and evaporate the combined extracts to dryness. Dissolve the residue in 0.2 mL of concentrated sulfuric acid and divide between two wells of a spotting tile. Add 0.1 mL of sodium nitrite solution to one well and 0.1 mL of chromotropic acid reagent to the other. Heat the tile at 80°C and observe any colour development.

Indications

See Table 30.9.

Table 30.9 Colours with sodium nitrite and chromotropic acid

Compound	Sodium nitrite	Chromotropic acid
2,4-Dichlorophenoxyacetic acid	Brown	Purple
2,4-Dichlorophenoxypropionic acid	Dark brown	Light purple
4-Chloro-2-methylphenoxyacetic acid	Light brown	Light purple
2-(4-Chloromethylphenoxy)propionic acid	Light brown	Purple
2,4,5-Trichlorophenoxyacetic acid	No reaction	Purple
2-(2,4,5-Trichlorophenoxy)propionic acid	No reaction	Light pink/purple

Cobalt thiocyanate (see Scott's test)

Reagent

1. A 2% (w/v) solution of cobalt thiocyanate in water
2. Phosphoric acid
3. 1 g $\text{H}_2\text{PtCl}_6 \cdot 6\text{H}_2\text{O}$ in 20 mL of H_3PO_4 .

Mix 9 parts of solution 1 and 3 parts of solution 2, add 1 part of solution 3 and mix well. Add 9 parts distilled water and mix. When the solution turns pink it is ready for use.

Method

Add a few drops of the reagent to the sample to be tested.

Indications

See Table 30.10.

Limit of detection (LOD): cocaine-HCl 60 µg, methadone-HCl 15 µg.

Table 30.10 Colours with cobalt thiocyanate

Colour	Compound
Blue (flaky precipitate)	Cocaine
Brilliant greenish blue	Benzfetamine-HCl, brompheniramine maleate, chlordiazepoxide-HCl, chlorpromazine-HCl, doxepin-HCl, hydrocodone tartrate, methadone-HCl, methylphenidate-HCl
Strong greenish blue	Diacetylmorphine-HCl, ephedrine-HCl, phencyclidine-HCl, procaine-HCl, propoxyphene-HCl, pseudoephedrine-HCl
Strong blue	Quinine-HCl

Coniferyl alcohol (primary aromatic amines)

Reagent

Warm 0.1 g of coniferyl alcohol until it melts (m.p. 74°C), dissolve in 3 mL of ethanol and dilute to 10 mL with ethanol.

Method

Place 1 drop of a solution of the sample on a filter paper, add 1 drop of the reagent and expose the paper to hydrochloric acid fumes.

Indications

An orange colour indicates the presence of an aromatic primary amine in which the amino group is attached directly to a benzene ring.

An anomalous reaction is obtained with diphenylamine (bright orange).

Copper sulfate

Method 1

Dissolve the sample in a minimum volume of 0.1 mol/L sodium hydroxide and add a 1% (w/v) solution of copper sulfate, drop by drop, until the colour change is complete.

Indications

Green, blue or brown colours indicate the presence of a sulfonamide (Table 30.11).

Table 30.11 Colours with copper sulfate, method 1

Colour	Compound
Green	Phthalylsulfathiazole, succinylsulfathiazole (→violet), sulfachlorpyridazine, sulfadimethoxine, sulfadimidine (→brown), sulfadoxine, sulfaethidole, sulfamerazine (→brown), sulfamethizole, sulfamethoxazole, sulfametopyrazine, sulfapyridine (→brown–green), sulfaquinoxaline, sulfasomidine
Blue	Phthalylsulfacetamide, sulfacetamide, sulfaguandine (→brown), sulfamethizole, sulfaurea, sulfinpyrazone, sulthiame
Brown	Sulfafurazole
Orange–brown	Sulfasalazine
Green–brown	Sulfamethoxypyridazine, sulfamoxole
Violet–brown	Sulfadiazine, sulfamethoxydiazine, sulfathiazole

Method 2

Add 1 or 2 drops of a 1% (w/v) solution of copper sulfate to the sample on a white tile.

Indications

A blue colour indicates the presence of an alkali salt of a fatty acid, such as sodium cromoglicate (1–2 min) or valproate. The colours are not produced by a change of pH (some of the alkali salts will change the pH), as negative results are obtained with sodium bicarbonate.

Cyanogen bromide**Reagent**

1. Decolorise bromine solution by the addition of solid potassium cyanide and then add more bromine solution until the solution is pale yellow.
2. Prepare a saturated solution of aniline in water.

Solutions 1 and 2 are stable for 1 week. Mix equal volumes of the two solutions immediately prior to the test.

Method

Add 1 drop of the mixed reagent to the sample on a white tile.

Indications

Red, orange or yellow colours indicate the presence of a mono-substituted pyridine ring. Increasing chain length of the substituent group weakens the response; a delayed response is obtained when the pyridine ring is substituted by nitrogen adjacent to the ring nitrogen; a weak response is obtained where there is a C=O substituent adjacent to the ring nitrogen. There is no response to the test if the pyridine ring is bound to another ring, if it is substituted in more than one position or if the nitrogen in the ring is substituted. Anomalous results are obtained with azatadine (pink), bisacodyl (no response) and tropicamide (violet–pink) (Table 30.12).

Diazotisation**Method**

Dissolve the sample in 2 mol/L hydrochloric acid, and to 1 drop on a white tile add 1 drop of a 1% solution of sodium nitrite, and 1 drop of a 4% solution of naphth-2-ol in 2 mol/L sodium hydroxide.

Indications

A bright red or orange–red colour indicates the presence of a primary aromatic amine. Diphenylamine does not give a reaction; amino–nitrothiazole (solid) gives a violet colour.

Dille–Koppanyi reagent modified (a general test for barbiturate-like compounds)**Reagent**

1. Dissolve 0.1 g of cobalt(II) acetate dihydrate in 100 mL of methanol. Add 0.2 mL of glacial acetic acid and mix.
2. Add 5 mL of isopropylamine to 95 mL of methanol.

Table 30.12 Colours with cyanogen bromide^(a)

Colour	Compound
Red	Zimeldine (30 s)
Pink	Azatadine
Orange–pink	Carbinoxamine, dimetindene, doxylamine, iproniazid, phenylamidol, triprolidine (1–2 min)
Violet–pink	Tropicamide
Orange	Azaperone, brompheniramine, chlorphenamine, isoniazid, metyrapone, nicametate, nicotinamide, nicotine, nicotinic acid, nifenazone, nikethamide, pheniramine, xanthinol nicotinate
Red–orange	Benzyl nicotinate
Yellow	Halopyramine, mepyramine, tripelenamine

^(a)Anomalous results are obtained with azatadine (pink), bisacodyl (no response) and tropicamide (violet–pink).

Method

Add 2 drops of solution 1 to the drug, followed by 1 drop of solution 2.

Indications

A light purple (blue–violet) colour indicates the presence of a barbiturate. Other reacting compounds are hydantoins, sulfonamides, pyrimidine, piperidine, methypylon. The LOD is 25 µg or lower.

p-Dimethylaminobenzaldehyde (Wasicky reagent or Van Urk reagent; a general test for ergot alkaloids)**Reagent**

Dissolve 2.0 g of p-dimethylaminobenzaldehyde (p-DMAB) in 50 mL of 95% ethanol and 50 mL of concentrated hydrochloric acid. The reagent should be freshly prepared.

Method

Add the reagent to the sample in a test-tube, warming if necessary. Observe any colour produced, then carefully dilute with water or spray dried spots on filter paper and heat.

Indications

Colours are given by a number of substances, which include ergot alkaloids, dimethyltryptamine, psilocin, psilocybine (gives a violet colour), cannabinols and certain indoles in which the indole ring is not bonded to another conjugated ring (red changing to violet on dilution), and certain phenols and phenolic amines (red or orange, usually changing to violet on dilution). Some other types of compound also respond. See Table 30.13.

The LOD for lysergide (LSD) is 6 µg.

Diphenylamine test**Reagent**

Mix 0.5 g of diphenylamine in 20 mL of water and dilute to 100 mL with concentrated sulfuric acid.

Method

Apply the reagent to the sample on a white tile or in a test-tube.

Indications

A blue colour indicates the presence of an oxidising agent such as bromate, chlorate, chromate, dichromate, iodate, lead(IV), manganese (III, IV, VII), nitrate, nitrite, permanganate or vanadate.

This test has been modified for use on blood samples to detect ethchlorvynol (Caughlin 1991). Blood (0.5 mL) is mixed with 1.0 mL of acetone and vortex mixed. The sample is centrifuged and 50 µL of the supernatant is added to 50 µL of diphenylamine reagent and 25 µL of chloroform. The mixture is vortex mixed and allowed to stand. A pink colour that develops in the chloroform layer indicates ethchlorvynol.

Dragendorff reagent (a general reagent for nitrogenous bases)**Reagent**

Dissolve 1 g of bismuth subnitrate in 3 mL of 10 mol/L hydrochloric acid with the aid of heat. Dilute to 20 mL with water and dissolve 1 g of

Table 30.13 Colours with p-dimethylaminobenzaldehyde

Colour	Compound
Red (changing to violet on dilution)	Cannabinols, phenazone (100°C, 5 min), pindolol, psilocin, psilocybine, tryptamine
Red (no violet on dilution)	Benserazide, cocaine (100°C, 3 min), feprazone, harmine, phenacyclidine (100°C, 3 min)
Orange (changing to violet on dilution)	Dobutamine, dopamine, diamorphine, morphine, orciprenaline, phenol, terbutaline, tyramine
Violet	Ergot alkaloids (dihydroergotamine, ergometrine, ergotamine, ergotamine, lysergide, methysergide), dimethyltryptamine, psilocin, psilocybine
Yellow	Primary aromatic amines, e.g. aminosalicilic acid, anileridine, aniline, procaine, benzocaine

potassium iodide in the mixture. If black bismuth triiodide separates, add 2 mol/L hydrochloric acid and more potassium iodide to dissolve it.

Method

Dissolve the sample in 3 drops of 2 mol/L hydrochloric acid, add 2–3 mL of the reagent and dilute to 10 mL with water.

Indications

An orange, red–orange or brown–orange precipitate suggests the presence of an alkaloidal base (precipitated as the alkaloidal bismuth iodide). Primary, secondary, tertiary and quaternary amines give positive results. This reagent is commonly used as a spray or locating agent to detect alkaloids on TLC plates.

Duquenois reagent, modified

Reagent

1. Add 2.5 mL of acetaldehyde and 2.0 g of vanillin to 100 mL of 95% ethanol
2. Concentrated hydrochloric acid
3. Chloroform.

Method

Place the solid sample, or an evaporated petroleum ether (or other organic solvent) extract of the sample, in a test-tube and add 3 drops of solution 1. Shake for 1 min and add 3 drops of solution 2. Agitate gently and observe the colour produced. Add 9 drops of solution 3, vortex mix gently and note whether the colour is extracted from the mixture.

Indications

A colour change from grey to green through blue to violet–blue suggests the presence of cannabis, but differentiation from roasted coffee and patchouli oil is required. The colour change is best seen with fresh drug material. The violet colour is extracted into the chloroform layer only when cannabis is present (Table 30.14). The LOD is 350 µg of tetrahydrocannabinol (THC). No colour is obtained with other natural products, such as basil, bay leaf, eucalyptus oil, mace, marjoram, rosemary, sage, thyme or tobacco.

Ferric chloride (general reagent for phenols, e.g. salicylates)

Reagent

Dissolve 5 g of anhydrous ferric chloride, or 8.25 g of ferric chloride hexahydrate, in 100 mL of distilled water.

Method

Add ferric chloride solution to the sample or an ethanolic solution of the sample.

Indications

Red, orange, green, blue, violet or brown colours indicate the presence of a phenolic compound, fatty acid or a phenylpyrazoline. High quantities of phenothiazines can also cause this test to be positive. Salicylates give a violet colour. Many phenols give no colour with ferric chloride when water is used as a solvent, but give positive tests when anhydrous solvents such as chloroform are used. Aspirin (acetylsalicylic acid) does not give a positive result unless first hydrolysed with concentrated sodium hydroxide to give salicylate. Colours are listed in Table 30.15.

Table 30.14 Colours with modified Duquenois reagent

Compound	Initial colour	Colour extracted by chloroform
Cannabis	Violet–blue	Violet
Coffee (roasted)	Violet–brown	Nil
Nutmeg	Pale reddish purple	Nil
Patchouli oil	Violet	Nil
Tea (leaves)	Green–blue	Nil

Table 30.15 Colours with ferric chloride

Colour	Compound
Red	Acetates, phenazone, propionates
Brown–red	Nifenazone
Orange	Chlorpromazine, hexoprenaline, propyphenazone, valproate
Yellow	
Green–yellow	Paracetamol
Green	Adrenaline, betanaphthol, dobutamine, dopamine, etamivan, ethylnoradrenaline, hexylresorcinol, hydroquinone, hydroxyquinoline, isoetarine, isoprenaline, levodopa, methylidopa, methylidopate, noradrenaline, paraphenylenediamine, phenothiazine, protokylol, rimiterol, thioridazine
Blue–green	Chlorquinaldol
Blue	Apomorphine, dodecyl gallate, gallic acid, morphine, parachlorophenol, pethidine, phenol, tannic acid
Violet	Aminosalicylic acid, diflunisal, dipyrone, hexachlorophene (transient), labetalol, salicylaldehyde, salicylamide, salicylic acid, salicyluric acid
Blue–violet	Aminophenazone, salicylamide (after hydrolysis), salicylic acid
Brown	Aloin, carbidopa
Yellow–brown	Salinazid
Green–brown	Benserazide
Black	
Violet–black	Ethyl gallate (→blue–black)

Ferrous sulfate A (test for nitrates and nitrites)

Reagent

To 1 volume of a 10% (w/v) solution of ferrous sulfate ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$) add 5 volumes of concentrated sulfuric acid with cooling.

Method

Add the sample to 0.5 mL of the reagent.

Indications

A red or pink colour is given only by nitrates and nitrites (e.g. glyceryl trinitrate).

Ferrous sulfate B (test for cyanide)

Reagent

Dissolve 10 g of ferrous sulfate in 100 mL of freshly boiled and cooled water (prepare fresh).

Method

Dilute 1 mL of sample with 2 mL of 10% (w/v) sodium hydroxide solution and add 2 mL of ferrous sulfate solution. Add sufficient 10% (v/v) hydrochloric acid to dissolve the ferrous hydroxide precipitate.

Indications

A blue colour is given by cyanide. There are no common sources of interference.

Folin–Ciocalteu reagent (test for phenolic compounds)

Reagent

For the stock solution, dissolve 100 g of sodium tungstate and 25 g of sodium molybdate in 800 mL of water in a 1500 mL flask, add 50 mL of phosphoric acid and 100 mL of concentrated hydrochloric acid, and reflux for 10 h. Cool, add 150 g of lithium sulfate, 50 mL of water and 4–6 drops of bromine, and allow to stand for 2 h. Boil for 15 min to remove the excess bromine, cool, filter and dilute to 1000 mL with water.

This stock solution should be stored at a temperature not exceeding 4°C and used within 4 months of its preparation; it has a yellow colour and must not be used if any trace of green colour is present.

For use, dilute 1 volume of this stock solution with 2 volumes of water.

Method

Add the diluted reagent to the sample and make the mixture alkaline with 2 mol/L sodium hydroxide.

Indications

A blue colour indicates the presence of a phenolic compound. The reaction is progressively inhibited with increased halogenation of the phenol nucleus.

Formaldehyde-sulfuric acid

Reagent

To 4 volumes of concentrated sulfuric acid add 6 volumes of formaldehyde solution (using a pipette with the tip below the surface of the acid) with stirring and adequate cooling. When the reagent is warm it remains clear for about 1 h. If turbidity develops, this may be dispelled by heating in a water-bath at 100°C for about 1 min (note that this reagent is not the same as that used in the Marquis test).

Method

Mix the sample with the reagent and heat at 100°C for 1 min.

Indications

Benzodiazepines generally give an orange colour with the exception of bromazepam and clozapine (a benzodiazepine-like compound), which both give yellow, and flurazepam (pink). Other indications include phenothiazines, tetracyclines and thioxanthenes. Tryptamine (brown) and zomepirac (red) also react. Those marked with an asterisk in Table 30.16 fluoresce orange under ultraviolet (UV) light ($\lambda = 350$ nm). No response is obtained with chlordiazepoxide, dimethoxanate or proquamezine. Some of the newer benzodiazepines have not been tested.

Table 30.16 Colours with formaldehyde-sulfuric acid

Colour	Compound ^(a)
Red	Chlorprothixene*, clopenthixol*, flupentixol*, fluphenazine, metopimazine, pericyazine, promazine, thiothixene, triflupromazine, zomepirac
Brown-red	Lymecycline, oxytetracycline, tolmetin
Pink	Flurazepam, thioproperazine, trifluoperazine
Orange	Clonazepam, clorazepic acid, demeclocycline (→brown-red), demoxepam, diazepam, flunitrazepam, ketazolam, lorazepam, lormetazepam, medazepam (add water), metixene, nitrazepam, nordazepam, oxazepam, prazepam, temazepam, tetrazepam
Red-orange	Methacycline
Yellow	Bromazepam, clozapine, dimethothiazine, doxycycline
Green-yellow	Rolitetracycline (→yellow-brown), tetracycline (→yellow-brown)
Green	Thiethylperazine
Blue	Carphenazine, levomepromazine, thioridazine
Violet	Mesoridazine, perphenazine
Red-violet	Alimemazine, chlorpromazine, diethazine, mequitazine, methdilazine, perazine, phenothiazine, prochlorperazine, profenamine, thiopropazate
Blue-violet	Acepromazine, acetophenazine, piperacetazine, promethazine, propiomazine
Brown-violet	Thiazinamium
Brown	Tryptamine
Orange-brown	Chlortetracycline, clomocycline

^(a)Compounds giving colours that fluoresce under UV light ($\lambda = 350$ nm) are indicated by an asterisk.

Table 30.17 Colours with Forrest reagent

Colour	Compound
Red	Acepromazine, carfenazine, chlorpromazine, diethazine, dimetotiazine, mequitazine, mescaline, mesoridazine, piperacetazine, prochlorperazine, promethazine, propiomazine, thiazinamium, thiopropazate, thioproperazine, thioridazine
Violet-red	Perphenazine
Brown-red	Alimemazine
Pink	Profenamine
Orange	Fluphenazine, phenothiazine, trifluoperazine, triflupromazine
Pink-orange	Acetophenazine
Red-orange	Methdilazine
Brown-orange	Perazine, pericyazine
Green	
Blue-green	Clomipramine, desipramine, imipramine, ketamine, opipramol, thiethylperazine, thioridazine, trimipramine
Violet	Levomepromazine, proquamezine (→red→orange)
Brown	Metopimazine
Red-brown	Promazine

Forrest reagent

Reagent

Mix together equal volumes of a 0.2% (w/v) solution of potassium dichromate, a 30% (v/v) solution of sulfuric acid, a 20% (w/w) solution of perchloric acid and a 50% (v/v) solution of nitric acid.

Method

Dissolve the sample in a minimum volume of 2 mol/L hydrochloric acid and add an equal volume of the reagent. To test urine, add 1 mL of reagent to 0.5 mL of urine.

Indications

Red, pink, orange, blue or violet colours are obtained with phenothiazines. A blue colour is obtained with certain dibenzazepines. The blue colour is inhibited by the presence of phenothiazines, so an excess of reagent must be added to overcome this. Colours are listed in Table 30.17.

FPN reagent (general reagent for phenothiazines)

Reagent

Mix together 5 mL of 5% (w/v) ferric chloride solution, 45 mL of a 20% (w/w) solution of perchloric acid and 50 mL of a 50% (v/v) solution of nitric acid.

Method

Dissolve the sample in a minimum volume of 2 mol/L hydrochloric acid (or use 1 mL of urine) and add an equal volume of the reagent.

Indications

A variety of colours, from pink, red, orange, violet to blue, indicate the presence of phenothiazines (Table 30.18).

Froehde reagent

Reagent

Dissolve 1.0 g of molybdic acid or sodium molybdate in 100 mL of hot concentrated sulfuric acid.

Method

Add a drop of the reagent to the sample on a white tile.

Indications

Colours are listed in Table 30.19.

Table 30.18 Colours with FPN reagent

Colour	Compound
Red	Chlorpromazine, dimetotiazine, mesoridazine, methdilazine, prochlorperazine, thiazinamium, thiopropazate
Orange-red	Mequitazine
Violet-red	Perphenazine, proquamezine (→red→orange)
Brown-red	Alimemazine
Orange	Acetophenazine, diethazine (→yellow), fluphenazine, metopimazine, morphine, pericyazine, phenothiazine, profenamine, promethazine, thioproperazine, trifluoperazine, triflupromazine
Red-orange	Carfenazine
Pink-orange	Propiomazine (→red→fades)
Brown-orange	Acepromazine, perazine, piperacetazine
Blue	Clomipramine, imipramine, thiethylperazine, thioridazine, trimipramine
Violet	Levomepromazine
Brown	
Red-brown	Promazine

Table 30.19 Colours with Froehde reagent

Colour	Compound
Yellow	Hydrocodone, pethidine
Blue-yellow	Oxycodone-HCl
Orange	Diphenhydramine, flurazepam, promazine, trifluoperazine, triflupromazine
Green	Chlorphentermine, codeine, mescaline, oxycodone, phenyltoloxamine
Yellow-green	Lysergide
Blue	Pentazocine
Red	Amphetamine, chlorpromazine-HCl
Grey-red	Propoxyphene-HCl
Purple-red	Alimemazine, diacetylmorphine, promethazine, propylhexadrine, salicylic acid, tetracycline, thioridazine
Brown	Ephedrine, mescaline
Red-brown	Doxepin-HCl
Black	
Brown-black	Opium
Green-black	Methylenedioxyamphetamine (MDA)-HCl

Fujiwara test (general reagent for halogenated hydrocarbons)

Reagent

Freshly prepared 20% (w/v) sodium hydroxide solution.

Method

Mix together 2 mL of the reagent and 1 mL of pyridine. Add the sample (1 mL of urine) and heat in a water-bath at 100°C for 2 min with shaking.

Indications

A red–pink colour in the pyridine layer indicates the presence of compounds that possess at least two halogen atoms bound to one carbon atom. These include chloramphenicol, chlorbutanol, chloroform, dichlorophenazone, trichloroethane, trichloroethanol, trichloroacetic acid and trichloroethylene. Chloral hydrate and dichlorophenazone do themselves react but are excreted in urine as trichloroacetic acid. No colour is given by dicophane (DDT) or carbon tetrachloride, although massive exposure to the latter solvent may lead to a positive urine test because of the presence of chloroform as a contaminant. 2,2,2-Trichloroethanol gives a yellow colour. The LOD is 1 mg/L.

Furfuraldehyde (general reagent for carbamates)

Reagent

A 10% (v/v) solution of furfuraldehyde in ethanol.

Method

Dissolve the sample in ethanol, place a drop of the solution on a filter paper, add 1 drop of the reagent and expose the paper to hydrochloric acid fumes for 2–3 min.

Indications

A black spot indicates the presence of non-aromatic carbamates. *N*-Substituted carbamates do not react. The LOD is 1 µg.

Iodine test

Method

Mix the sample with an equal volume of manganese dioxide and heat the mixture carefully to dull redness over a small flame. Repeat the test by heating the sample alone.

Indications

The appearance of violet vapour indicates the presence of iodine in the molecule. Better results are sometimes obtained when the manganese dioxide is omitted (e.g. with amiodarone).

Iodoplatinate test (general test for alkaloids and nitrogenous heterocyclic compounds)

Reagent

Add 2 mL of a 5% (w/v) solution of platonic chloride and 5 g of potassium iodide to 98 mL of water and shake until dissolved. This reagent is often used as a locating agent in TLC.

Method

Dissolve the sample in 2 drops of 2 mol/L hydrochloric acid, add 2–3 mL of the reagent and dilute to 10 mL with water.

Indications

A violet, blue–violet, brown–violet or grey–violet precipitate suggests the presence of an alkaloidal base (precipitated as the alkaloid–iodoplatinate complex). The clearest colours are obtained with tertiary and quaternary amines; primary amines give indistinct colours and amines of small relative molecular mass generally do not react.

Koppanyi-Zwicker test

Reagent

A 1% (w/v) solution of cobalt nitrate in ethanol.

Method

Dissolve the sample in 1 mL of ethanol, add 1 drop of the reagent followed by 10 µL of pyrrolidine and agitate the mixture.

Indications

A violet colour is given by substances that contain the following structures:

- Imides, in which C=O and NH are adjacent in a ring (e.g. barbiturates, glutethimide, oxyphenisatine and saccharin).
- Sulfonamides and other compounds with free –SO₂NH₂ on a ring (e.g. clopamide, furosemide, sulfanilamide and thiazides), or with –SO₂NH₂ in a side-chain (e.g. chlorpropamide), or with –SO₂NH₂ that links a benzene ring with another ring other than a pyrazine, pyridazine, pyridine or pyrimidine ring (e.g. sulfafurazole and sulfamethoxazole). These latter structures give pink or red–violet colours (e.g. sulfadiazine and sulfadimethoxine).

No response is obtained with compounds with other substituents on the nitrogen atom. Anomalous responses are obtained with paramethadione and theophylline (violet), and with cycloserine, idoxuridine, mephentyoin, niridazole and riboflavin (no response). Note that hydrochlorides give a blue colour before the addition of pyrrolidine.

Liebermann's reagent**Reagent**

Add 1 g of sodium or potassium nitrite to 10 mL of concentrated sulfuric acid with cooling and swirling to absorb the brown fumes.

Method

Add 2 or 3 drops of the reagent to the sample on a white tile. Occasionally it is necessary to carry out the test in a tube and heat in a water-bath at 100°C. Many substances give colours with sulfuric acid alone and the test should be repeated using sulfuric acid instead of the reagent.

Indications

This test was originally developed to give intense colours with phenols:

- Orange colours are given by substances that contain a monosubstituted benzene ring not joined to C=O, N-C(=O)- or to a ring that contains a C=N-O- group.
- Orange or brown colours are given by some substances that contain two monosubstituted benzene rings (or some disubstituted compounds in which fluorine is the second substituent) that are joined either to one carbon atom or to adjacent carbon atoms.
- A wide range of colours is given by compounds that contain -OH, O-alkyl or -O-CH₂O- groups attached to a benzene ring or to a ring in a polycyclic structure that contains a benzene ring. The benzene ring must not bear -NO₂, or be halogenated, or contain an -O- substituent *ortho* to the oxy groups. Compounds that contain ring sulfur give a similar range of colours.

Colours are listed in Table 30.20. Note that a yellow colour is given by a variety of other compounds.

Mandelin's test (useful test for amfetamines and antidepressants)**Reagent**

Dissolve 1.0 g of ammonium vanadate in 1.5 mL of water and dilute to 100 mL with concentrated sulfuric acid.

Method

Add a drop of the reagent to the sample on a white tile.

Indications

When interpreting the result of this test, account should be taken of the colour given by sulfuric acid and by Liebermann's test. Hydrochlorides give a red colour with this reagent. When the colours differ from those given with sulfuric acid or Liebermann's test, this indicates an aromatic ring together with a saturated 5-, 6- or 7-membered ring that contains only one nitrogen atom. The heterocyclic ring must not contain a second nitrogen atom or an oxygen atom. It must not be substituted or bound by -CONH- to the aromatic ring. The aromatic ring must not have -CF₃ as a substituent. Colours are also produced if sulfur is in a ring, provided that the ring does not contain more than one nitrogen atom (Table 30.21).

LOD values are: codeine sulfate 5.0 µg, amphetamine-HCl 10.0 µg, diamorphine-HCl 20 µg, metamfetamine 150 µg, morphine 5 µg and strychnine 0.05 µg.

Marquis test

The Marquis test is a useful broad-spectrum test used mostly for opium alkaloids and amfetamines.

Reagent

Carefully mix 100 mL of concentrated sulfuric acid with 1 mL of 40% (v/v) formaldehyde solution (stable for several weeks if protected from light).

Method

Add a drop of the reagent to the sample on a white tile.

Indications

Various colours that represent the whole of the visible spectrum are given by a large number of compounds. Structures that tend to maintain

the response to the reagent at the violet end of the spectrum are, in decreasing order of efficacy: ring sulfur (with or without aromatic ring); ring oxygen (with aromatic ring); extra-ring oxygen or sulfur (with aromatic ring); aromatic compounds that consist entirely of C, H, N. Thus, there is a tendency for the response to the Marquis reagent to move gradually towards longer wavelength (i.e. through green to orange and red) as the ratio of C, H, N to the other groups in the molecule rises (Table 30.22).

The LOD values are: 1 µg for codeine sulfate, mescaline sulfate, methadone-HCl; 5 µg for lysergide tartrate, metamfetamine-HCl and morphine; 10 µg for amphetamine-HCl and diamorphine-HCl.

McNally's test**Reagents**

1. A 0.5% solution of copper sulfate in 10% acetic acid.
2. A freshly prepared 2% (w/v) solution of sodium nitrite.

Method

Dissolve the sample (1 mg) in a few drops of acetone, and add 1–2 mL of water. Add 3 drops of solution 1 and an equal volume of solution 2. Shake and heat in a water-bath at 100°C for 3 min.

Indications

A red colour indicates the presence of free salicylic acid. Aminosalicic acid gives a brown precipitate, and diflunisal gives a violet colour. Certain acids produced during the putrefaction of tissues also give red colours in this test: *p*-hydroxyphenylacetic acid, *p*-hydroxyphenylpropionic acid and *p*-hydroxyphenyl-lactic acid.

Mecke's reagent (useful test for opium alkaloids)**Reagent**

Dissolve 1.0 g of selenious acid in 100 mL of concentrated sulfuric acid.

Method

Add a drop of the reagent to the sample on a white tile.

Indications

An immediate blue or green colour is indicative of opiates (see Table 30.23).

Melzer's reagent (general reagent for hallucinogenic mushrooms)**Reagent**

Dissolve 1.5 g of iodine in 100 mL of an aqueous solution that contains 5 g of potassium iodide and 100 g of chloral hydrate.

Method

Place a few drops of the reagent on the mushroom spores or mushroom tissue to be tested.

Indications

A blue, bluish-grey or black-grey colour indicates amyloid mushrooms. A slight yellow or no change indicates that the mushrooms are non-amyloid. Psilocybes are always non-amyloid.

Mercurous nitrate (general reagent for barbiturate-like compounds)**Reagent**

To a saturated solution of mercurous nitrate, add solid sodium bicarbonate until effervescence ceases and the precipitate formed becomes yellow. The precipitate then changes to a biscuit colour. This reagent should be freshly prepared and should be shaken immediately before use, and should not be kept for more than 1 h.

Method

Dissolve the sample in the minimum amount of ethanol, add 1 drop of the opaque reagent, shake and examine at intervals during 2 min. A blank solution that contains only ethanol and reagent should be treated similarly at the same time.

Table 30.20 Colours with Liebermann's reagent

Colour	Compound
Red	Acepromazine, ajmaline, alprenolol, aminacrine (100°C), antazoline, brucine, chlorprothixene, clopenthixol, flupentixol, mestranol, oxytetracycline, prajmalium, thiazinamium, tiotixene, tolmetin (100°C), trifluoperazine, xylazine
Violet-red	Indapamide
Brown-red	Methylchlorophenoxyacetic acid
Pink	Trichlorophenoxyacetic acid (→brown)
Brown-pink	Prazosin (100°C →red-orange)
Orange	Aletamine, alverine, ampicillin, atropine methobromide, atropine methonitrate, baclofen, benactyzine (→brown), bethanidine (→brown), broxyquinoline, butanilicaine, chloroquine (100°C), clidinium (→brown), cyclandelate, cyclizine, dazomet, decoquinat (slow), diethylthiambutene (100°C), dimeflin, diuron, doxapram, dyclonine (100°C), fenclofenac (100°C, →brown), fenitrothion, fenpipramide, glibenclamide (100°C, 15 s), hyoscine butylbromide, hyoscine methonitrate, linuron, loxapine (50–60°C), metindizate (→brown), methylphenidate, metolazone (→green-brown), monolinuron, nomifensine, phenazone (100°C), phenelzine, prophan, salinazid, sulfipyrazone, tolazoline, trimetaphan, tripeleminamine (→brown), triprolidine, xipamide, zomepirac (100°C)
Red-orange	Acetanilide, amfetamines, aniline, atropine, bamipine, beclamide, benethamine, caramiphen, carbetapentane, chlorcyclizine, cinchophen, cycrimine, diphenylpyraline, doxylamine, dropropizine, ephedrine, famprofazone, fencamfamin, glutethimide, hyoscine, hyoscyamine, isoaminile, isocarboxazid, levamisole, meclozine, mephentermine, methoin, methyl benzoate, methylphenobarbital, metixene, metomidate, morazone, nialamide, pentapiperide, pethidine, phenacetamide, phenbutrazate, phendimetrazine, phenglutarimide, pheniramine, phenmetrazine, phenobarbital, phensuximide, phenylmethylbarbituric acid, phenytoin, prolintane, tofenacin, tranlycypromine, triamterene, triphenyltetrazolium, warfarin
Brown-orange	Ambutonium, bumetanide, diphenhydramine, fenuron, feprazone (100°C, →brown), ibuprofen, labetalol, mepivacaine, methadone, nefopam (→brown), tetrahydrozoline
Yellow	Amicarbalide (100°C), clonidine (100°C, →orange), dequalinium (100°C, →orange), diethylpropion, diloxanide, ethoxzolamide, fenfluramine (100°C), flavoxate, gliclazide, metoclopramide, nifenazone (100°C), piroxicam, propachlor, tropicamide
Brown-yellow	Amiodarone
Green	Bialamicol, chlorotrianisene, colchicine, dextromoramide (100°C), diamthazole, hydrastine, mequitazine, naphthols, phenol, phenothiazine, thiocarlde
Blue-green	Hydrochlorothiazide, hydroflumethiazide, pindolol
Brown-green	Cyclopenthiiazide
Grey-green	Azapropazone
Black-green	Naproxen
Blue	Amidopyrine (100°C), bendroflumethiazide, benzonatate (100°C), chromonar (100°C, 3 min), clomipramine, diphenylamine, dipyrone (100°C), imipramine, mefenamic acid, mefruside, oxypertine, padimate (100°C), procarbazine (100°C, 15 s), propyphenazone (100°C; red with water), tetracaine (100°C), yohimbine
Green-blue	Amiphenazole (100°C)
Violet	Methocarbamol, mianserin, paracetamol, penthienate methobromide, phenacetin, propiomazine, resorcinol, timolol (100°C), trazodone (100°C; transient)
Red-violet	Chloroxuron
Black-violet	Methoxychlor
Brown	Acepromazine, acetophenazine, adiphenine, azacyclonol, barban, benzilonium, benzyl nicotinate, biperiden, clemastine, clofenotane, clomifene, cyclothiazide, dextropropoxyphene, dichlorprop, diperonon, diphepanil, difenidol, emepronium, etenzamide, fenpiprane, flurbiprofen, haloperidol, mepenzolate, methylpiperidyl benzilate, mexiletine, nadolol, penfluridol, phenaglycodol, phenylbutazone (100°C), phosalone, pimozide, pipazethate (100°C, →red), pipoxolan, pyrrobutamine, rotenone, sotalol (100°C), sulindac, veratrine, zimeldine
Red-brown	Benzthiazide, bisacodyl, carfenazine, chlorpromazine, diclofenac, dosulepin, profenamine, etisazole, fenbufen, fenoprofen, methapyrilene, perphenazine, polythiazide
Pink-brown	Metoprolol
Orange-brown	Benazolin, diphenadione, maprotiline, methiocarb, piperidolate
Green-brown	Methdilazine, norbormide, promazine, thiopropazate
Violet-brown	Bamethan, clofibrate, dichlorophen
Black-brown	Mecoprop
Grey	Isopropamide
Black	Acetomenaphthone, aloin, aminophenols, amodiaquine, apomorphine, atenolol, benorilate, benzquinamide, buprenorphine, butorphanol, carbaryl (→green), carbidopa, cephaeline, chloroxylenol, chlorphenesin, clomocycline, clorgyline, codeine, cotarnine, cresol, cyclazocine, dextromethorphan, diamorphine, dibromopropamidine, diprenorphine, doxepin, emetine, ethamivan, ethinylestradiol, estradiol, estriol, estrone, etilefrine, furosemide, glycopyrronium, guaiphenesin, hexobendine, hydroxyephedrine, hydroxystilbamidine, ibogaine, indometacin, levallorphan, mebeverine, mescaline, methylchlorophenoxyacetic acid, methylenedioxymfetamine, morantel, morphine, naloxone, 1-naphthylacetic acid, narceine, nicergoline, normetanephine, noscapine, noxiptiline, octafonium, oxprenolol, oxyphenisatine, papaverine, pholcodine, pizotifen, practolol, profadol, propanidid, protokylol, pyrantel, rimiterol, ritodrine, rotenone, salbutamol, terbutaline, tetrabenazine, tetracycline, thymol, trimethobenzamide, trimetozine, tubocurarine, verapamil, viloxazine

Table 30.21 Colours with Mandelin's test

Colour	Compound
Red	Ajmaline, amfetamine, azacyclonol, chlorprothixene, diperodon (→green), dofamium (→brown), flupentixol, gelsemine (→green), indapamide, mequitazine, methotrexate, nialamide, pericyazine, prajmalium, prolintane, sodium cromoglicate, tiotixene, xylometazoline
Brown-red	Diacetylmorphine-HCl, doxepin-HCl, nadolol, propoxyphene-HCl
Orange	Brompheniramine, dropropizine (slow), ethylnoradrenaline, hydrastinine (→green), lachesine (→green), levamisole (→grey-green), methanthelinium, metixene, methyl dopa, methyl dopate, methylpiperidyl benzilate (→brown→green), noradrenaline, orphenadrine, pipenzolate (→green), poldine metilsulfate (→green→violet), procaine-HCl, propantheline, proquamezine (→violet), solanidine (→violet→blue), solanine (→violet→blue), strychnine (blue→purple→violet→red→red-orange), sulindac, thenalidine (→brown)
Red-orange	Cotarnine (→brown), doxepin
Green-orange	5-Methyltryptamine
Brown-orange	Mexiletine
Yellow	Azaperone, benztropine, broxaldine, chelidonine (→green), conessine, dectropine, desipramine (→blue), dihydralazine, diphenhydramine, difenidol, diphenylpyraline, dropropizine (→orange), halquinol, homidium, lidoflazine, methacycline (→orange→violet), paraphenylenediamine, penicillamine, protokylol (→brown), tofenacin, tylosin (→yellow-brown), veratrine (→orange→violet→brown), viprynum
Orange-yellow	Cocaine-HCl, hexoprenaline, methaqualone, methylphenidate-HCl
Green-yellow	Methoxamine, oxycodone-HCl
Yellow-brown	Mescaline-HCl
Green	Acepromazine (→red), adiphenine (→blue), amfetamine, benorilate, bephenium hydroxynaphthoate, bibenzonium, buclosamide (blue rim), bunamidine, chlorpromazine (→violet), clefamide (→brown), codeine (green→blue), colchicine, cyclazocine, cyclomethycaine (→brown), debrisoquine, diaveridine, dibenzepin, diethazine (→violet with excess reagent), diethylthiambutene (→green→blue), dimethindene, dimethoxanate (→brown), dimoxylone, dipipanone (→blue), dosulepin, doxorubicin, doxycycline (→yellow), fenpiprane, guanoxan, harman, hydroxyephedrine, isoxsuprine, metanephine, methadone (→blue), methdilazine (→violet), methocarbamol, methoxyamfetamine, methylenedioxyamfetamine (→blue), α-methyltryptamine (→orange), metopimazine, monocrotaline, niclosamide, nitroxoline, norharman (→yellow), normetanephine, obidoxime (→blue), oleandomycin, oxymetazoline, paracetamol, pecazine (→violet), pentazocine, perazine (→violet), phenazone, phenazopyridine, phenformin, phenindamine, phenoxybenzamine (→violet), phenyltoloxamine, pindolol, piperacetazine (→red→violet), pipoxolan (→brown), prenylamine, profenamine (→violet), proflavine, promazine (→violet), promethazine (→violet), propranolol, reserpine, ritodrine, salicylic acid, thenium, thenyldiamine, thiocarlide (→yellow), tranlycypromine (→violet), trifluomeprazine (→red→violet), trihexyphenidyl
Yellow-green	Benzfetamine-HCl, metamfetamine-HCl, normethadone, opipramol
Blue-green	Amfetamine-HCl, benzoctamine, berberine (→brown), edrophonium, hydroxystilbamidine, ketobemidone, methoxyphenamine, phentolamine, profadol (→green), viloxazine
Brown-green	Benzylamine, chlorphenesin
Grey-green	Alverine, azapropazone, diamphenethide, diethyltryptamine (→yellow), dihydrocodeine, guaifenesin, hordenine, levomethadyl acetate, normorphine, oxyphenyclimine, papaverine, terbutaline, trihexyphenidyl
Blue	Bamethan (→green), clomipramine, deserpidine (→green), desferrioxamine (→violet), doxapram, droperidol (→green), harmine (→green), imipramine (add water), maprotiline, mebhrolin, metaraminol, phenaglycodol, phenylamidol, pyridoxine (→grey-green), salbutamol (blue rim→brown rim), strychnine (blue→purple→magenta→red→red-orange→orange), thioridazine (→violet), trimipramine (add water), triphenyltetrazolium (slow), xipamide, xylazine, yohimbine (→green)
Green-blue	Chlophedianol, labetalol
Violet	Alimemazine, amidefrine, benperidol, bezitramide (→orange), bisacodyl, captodiame, cefaloridine, chloropyrilene (→orange), clomifene (→orange→brown), clomocycline (→brown), denatonium, dipyridamole, guanoclor (→orange→brown→yellow), guanoxan, hexobendine, hydromorphone (→orange), mepacrine (→yellow), mepyramine, metisazone (→yellow), mianserin, morantel, naloxone (→brown), oxyclozanide (→orange), oxyphenisatine, oxytetracycline (→red→orange), penthienate, perphenazine, phenylbutazone, pizotifen (→green), prilocaine, primaquine (→orange), propiomazine, pyrantel, pyrobutamine, rolitetracycline (→red→orange), tetracycline (→red→orange), thiethylperazine, thiopropazate, triacetyloleandomycin (slow), tridihexethyl, trimetazidine
Red-violet	Antazoline, carfenazine, dimethothiazine, histapyrrodine, thonzylamine
Blue-violet	Alcuronium, hexocyclium, levomepromazine
Brown-violet	Alprenolol, bitoscanate, butaperazine, naphazoline
Grey-violet	Methadone-HCl, methoserpidine, oxprenolol, tricyclamol
Black-violet	Methylenedioxyamfetamine-HCl, methapyrilene
Brown	Amitriptyline (→green), azapetine, bampine, carbetapentane (slow), clidinium (→green), cyclopentolate, dipheanil, dipyrone, doxepin, embutramide, fluanisone, fluphenazine, isoeptarine, isometheptene, isoprenaline, metindizate, methyl benzoquate, methysergide, metoclopramide, norpiperanone (→blue), nortriptyline (→green), opium, phenelzine, phenylephrine, pimozide, piperidolate, prochlorperazine (→violet), propoxycaine, rescinnamine, salinazid, stanozolol, tetrabenazine, thioproperazine (→green→violet), tolafate, tolpropamine, tramazoline, tubocurarine
Red-brown	Benzthiazide, clioxanide, cycrimine, decoquinat, diclofenac, ethomoxane, hydrastine (→red), trifluoperazine, triflupromazine
Pink-brown	Metoprolol
Orange-brown	Rifampicin, spiramycin, thebaine
Yellow-brown	Clemastine, clofazimine, physostigmine, rifamycin SV, trimethoprim, tripelennamine
Green-brown	Etenzamide, harmaline, lysergic acid, mesoridazine, narceine, pentazocine, phenyltoloxamine, syrosingopine
Violet-brown	Chlortetracycline (→yellow), cyproheptadine, demeclocycline, dihydroergotamine, ergotamine, lymecycline (→yellow), methylergometrine, nicergoline (→brown), octaphonium, oxethazaine, protriptyline, trimethobenzamide
Grey-brown	Dextropropoxyphene, mephenesin carbamate

Table 30.21 continued

Colour	Compound
Grey	Dihydromorphine, diprenorphine, etilefrine (→green→brown), ibogaine (→violet), indometacin, lobeline, lysergide, morphine, oxypertine, propranolol, trazodone (→violet)
Blue-grey	Alphaprodine, diamorphine, morphine
Black	Procyclidine
Grey-black	Flurazepam

Table 30.22 Colours with the Marquis test

Colour	Compound
Red	Alprenolol, benzylmorphine (→violet), buphenine, dimethothiazine, etenzamide, etilefrine, fenclofenac (slow), fenpiprane, fluphenazine, flurbiprofen, hexoprenaline, labetalol (→brown-red), maprotiline, mephensin carbamate, mequitazine (slow), mesoridazine (→violet), methoxyphenamine, metopimazine, mexiletine, nadolol, pentazocine (→green), pericyazine, phenazopyridine, phenoperidine, phenylephrine, piperacetazine, prenylamine, thebaine (→orange), thiethylperazine (→green), thioproperazine, tiotixene, tolpropamine, tranlycypromine (→brown), vinblastine
Orange-red	Alverine, amfetamine-HCl, bethanidine, diphehanil, flupentixol, metamfetamine-HCl
Violet-red	Thioridazine (→blue-green)
Black-red	Doxepin-HCl
Brown-red	Alphaprodine, doxepin, trihexyphenidyl
Pink	Alimemazine, fenoprofen, fluopromazine, metoprolol, promazine, promethazine, trifluoperazine
Orange	Adrenaline (→violet), aletamine, amfetamine (→red→brown), anileridine (slow), benactyzine (→green→blue), benzethonium, benzilium (→green→blue), benzfetamine, bunamidine (→red), carbetapentane (slow), carfenazine (→red-violet), chlorphentermine, clidinium (→blue), cyclandelate (slow), cycrimine (→red), dehydroemetine, dimethyltryptamine, dipyridamole, ethacridine (→red), ethoheptazine, ethylnoradrenaline (→brown), famprofazone, fenbufen (→brown), fencamfamin, fenethylline, fentanyl, harmine, indapamide (→violet), indometacin, isothipendyl, ketobemidone, lachesine (→green→blue), lymecycline, mepenzolate (transient), mephentermine (→brown), mescaline, metamfetamine, metanephine (→violet-brown), methacycline, methanthelinium, methindizate (→green), methylphenidate, methylpiperidyl benzilate (→green→blue), 5-methyltryptamine (→brown), α -methyltryptamine (→brown), N-methyltryptamine, nefopam (→brown), nomifensine (slow), normetanephine (→violet-brown), oxeladin, oxytetracycline, pentapiperide, pethidine, phenethylamine, phenformin (→brown), phentermine, piminodine, pipenzolate (→green→blue), piperidolate, pizotifen (→red), poldine methylsulfate (→green→blue), primaquine, profadol (→red-brown), prolintane (→brown), propanteline, prothipendyl, psilocybine, rolitetracycline, spiramycin, tetracycline, trimethoprim, trimethoxyamfetamine, tryptamine, veratrine, xylometazoline
Red-orange	Chlorprothixene
Pink-orange	Diuron
Yellow-orange	Orphenadrine, pipradrol
Brown-orange	Amitriptyline
Yellow	Acriflavine (→red), amiloride, azacyclonol, benzquinamide, benzatropine, bromazine, broxaldine, broxyquinoline, caramiphen, chlordiazepoxide, chlorphenoxamine (→green), chlortetracycline (→green), chlortalidone, cinchophen, clemamide, clemastine (green rim), colchicine, conessine (→orange), cyclizine, demeclocycline (→green), depropine, diethyltryptamine (→brown), 2,5-dimethoxy-4-methylamfetamine, diphenhydramine, diphenidol, diphenylpyraline, doxycycline, ethoxzolamide, ethylmorphine (→violet→black), furaltidone, halquinol, hydrocodone (→brown→violet), hydromorphone (→red→violet), hydroxyephedrine, isoetarine (→orange), lidoflazine, lorazepam, mepacrine, methylidopa (→violet), methylidopate (→violet), norcodeine (→violet), orciprenaline, oxycodone (→brown→violet), oxyphenbutazone, phanquone, phenbutrazate (slow), phentolamine, phenylramidol, pindolol (→brown), pramoxine (→green), proflavine (→orange), salbutamol, salinazid, sodium cromoglicate, solanine (→violet), terbutaline, tetrabenazine, thebacon (→violet), tofenacin, triamterene, trimetazidine (fades), vancomycin, viprynum embonate, zomepirac (100°C, →orange)
Orange-yellow	Methylphenidate-HCl, stanozolol
Pink-yellow	Methadone-HCl
Green	Berberine, carbaryl, chelidonine, harman, norharman, oleandomycin, propranolol, protriptyline, pseudomorphine, sulindac (slow)
Yellow-green	Acepromazine (→red), verapamil (→grey)
Blue-green	Tolnaftate
Brown-green	Harmaline
Grey-green	Cyproheptadine, deserpidine, naphazoline, oxypertine, phenindamine, protokylol, rescinnamine, reserpine (→brown)
Blue	Clofibrate, embutramide, nicergoline (→grey)
Grey-blue	Mebhydrolin, 1-naphthylacetic acid
Violet	Alimemazine, apomorphine (→black), azatadine, benorilate, bisacodyl, buprenorphine, butriptyline, captodiame, chloropyrilene, chlorpromazine, clofazimine, codeine, diamorphine, diethylthiambutene, dihydrocodeine, dimethindene (→blue), dimethoxanate, doxorubicin, doxylamine, etoxazene, guaifenesin, guanoxan, hexocyclium metilsulfate, mepyramine, 6-monoacetylmorphine, morphine, nalorphine, normorphine, oxprenolol, oxycodone-HCl, oxyphenisatine, pecazine, penthienate, pentazocine, perazine, perphenazine, phenoxybenzamine, phenyltoloxamine, pholcodine, pimozone, pipoxolan (→grey), prochlorperazine, procyclidine, profenamine, promazine, promethazine, proquamezine, solanidine, thenium, thiopropazate, tricyclamol, viloxazine
Red-violet	Acetophenazine, benzoctamine, bephenium hydroxynaphthoate, cefaloridine, chlophedianol (→brown), dihydromorphine, ethomoxane, isoxsuprine, lobeline, methdilazine, propiomazine, tramazoline, trifluomeprazine, trifluoperazine, triflupromazine, trimiperidine
Blue-violet	Methocarbamol, levomepromazine, morantel, neopine, noscapine (fades), pyrantel

table continued

Table 30.22 continued

Colour	Compound
Brown-violet	Butaperazine, dopamine, methylenedioxyamphetamine, tridihexethyl
Grey-violet	Diprenorphine, oxymorphone, pyrrobutamine, thenalidine, trihexyphenidyl
Black-violet	Dextropropoxyphene (→green), methapyrilene, thenyldiamine
Brown	Bibenzonium, carbidopa, cyclazocine (→green), diclofenac (slow), dimoxyline, dosulepin, doxepin, ergometrine, ergotamine, erythromycin, hordenine (→green), ibuprofen (100°C, →orange), isoprenaline (→violet), lysergamide, methadone, naloxone (→violet), naproxen, narceine (→green), noradrenaline, pethidine-HCl, phentermine, phenazocine, rimiterol (→black), serotonin (slow), syringopine, tyramine (→green)
Red-brown	Biperiden, debrisoquine, methyl benzoate, oxetacaine, phenprobamate, trimetozine, tripeleminamine
Orange-brown	Benethamine (→brown), clomocycline, nortriptyline
Yellow-brown	Moxisylyte, ritodrine, triacetyloleandomycin, tylosin
Green-brown	Alcuronium, bufotenine, psilocin
Violet-brown	Clomifene, diethazine, levomethadyl acetate (→grey-green), methoxamine (→green)
Grey-brown	Dihydroergotamine, methylergometrine, octafonium
Grey	Butorphanol, diaveridine (→violet-brown), ibogaine (→orange), lysergide, methoserpidine, methysergide, pholedrine (→green)
Blue-grey	Acetorphine (→yellow-brown), etorphine (→yellow-brown)
Black	Methylenedioxyamphetamine
Blue-black	Methylenedioxyamphetamine
Green-black	Lysergide

Table 30.23 Colours with Mecke's reagent

Colour	Compound
Green	Diacetylmorphine, mescaline-HCl, morphine, oxycodone-HCl
Blue-green	Codeine, diacetylmorphine-HCl, hydrocodone tartrate, methylenedioxyamphetamine-HCl
Brown-green	Methadone
Orange	Alimemazine, diphenhydramine, fluopromazine, pethidine, phenyltoloxamine, promazine, promethazine, propoxyphene, trifluoperazine, trifluorpromazine
Yellow	Amphetamine, procaine
Red	Doxepin-HCl
Black-red	Chlorpromazine-HCl
Purple-red	Tetracycline
Violet (dark blue)	Methylenedioxyamphetamine, thioridazine
Brown	Ephedrine
Red-brown	Propoxyphene-HCl
Black	
Green-black	Lysergide, opium

Indications

A dark grey or black colour indicates a ring imide group or sulfonamides with an additional ring. The speed and intensity of the reaction varies between different compounds. The following ring imides react in decreasing order of intensity: barbiturates, bemegride, phenytoin > benperidol, cycloserine, pimozone > glutethimide, oxyphenisatine > saccharin, sulfapyrazole. In the case of sulfonamides, succinylsulfathiazole, sulfamoxole, sulfanilamide, sulfasomidine and sulfathiazole react with greater intensity than all others. Chlorpropamide and tolbutamide give a moderate response. If used as a spray, the LOD is 1–5 µg for barbiturates.

Methanolic potassium hydroxide**Reagent**

A 20% (w/v) solution of potassium hydroxide in methanol.

Method

Add a few drops of the reagent to a solution of the sample in methanol and heat if necessary to boiling point to develop the colour.

Indications

A change from colourless or from a pale colour to red, orange, yellow, green or blue is given by quinones, diones that possess an aromatic ring, phenols with adjacent hydroxy groups and compounds that contain nitro groups on a ring (Table 30.24). Many of these compounds are coloured already and give pale or colourless solutions in methanol.

Millon's reagent (general reagent for phenols)**Reagent**

Dissolve 3 mL of mercury in 27 mL of fuming nitric acid and add an equal volume of water with stirring.

Method

Add 0.5 mL of reagent to the sample and warm the mixture.

Table 30.24 Colours with methanolic potassium hydroxide

Colour	Compound
Red	Benserazide, isoetarine (→orange→yellow), metronidazole, nitrofurazone, phenindione
Orange-red	Fenitrothion, quintozone, tecnazene, trifluralin
Pink	Levodopa (→red-brown)
Orange-pink	Rimiterol
Brown-pink	Dobutamine
Orange	Aciniazole, barban, carbidopa, dinitro-orthocresol (100°C), dinobuton, dinoseb, dodecyl gallate, hexoprenaline (→brown), isoprenaline (→yellow), nifedipine, nifuratel, nifursol, obidoxime, protokylol (→yellow)
Pink-orange	Adrenaline(→brown)
Yellow-orange	Nitrofurantoin
Yellow	Acebutolol, diphenadione, methylidopa (→orange), metolazone, niclosamide, niridazole, nitroxoline, nitroxylin, phanquone (→brown-violet), sodium cromoglicate
Green-yellow	Phytomenadione (→violet→brown)
Green	Apomorphine (→red), dinitolmide
Blue	Dopamine (→orange→brown), methylidopate (→orange), noradrenaline (→orange)
Violet	Dimetridazole (when boiled)

Indications

A red or orange-red colour indicates the presence of a phenolic substance. Primary aryl amines also react. Some basic compounds that contain a phenolic group do not react to this test; a combination of this test with the Folin–Ciocalteu reagent is therefore advised for phenolic compounds. Phenols that contain more than one hydroxyl group do not give the typical red colour. This reagent does not react with phenols substituted with Cl, Br or I.

Naphthol-sulfuric acid

This test should be carried out in conjunction with the sulfuric acid test.

Reagent

Mix 1 g of naphth-2-ol with 40 mL of concentrated sulfuric acid and heat in a water-bath at 100°C, with occasional stirring, until the naphth-2-ol is dissolved.

Method

Mix the sample with 1 mL of the reagent, heat in a water-bath at 100°C for 2 min and note any colour produced. Cool, add 1 mL of water and note the colour again.

Indications

A range of colours is obtained with steroidal structures (Table 30.25). A positive response to this test combined with a positive response to the sulfuric acid test is indicative of the presence of a steroid.

Compounds other than steroids that give colours with this test include chloral hydrate and chloramphenicol (brown–yellow), starch and tartaric acid (green).

Nessler's reagent**Reagent**

1. Dissolve 50 g of mercuric chloride and 35 g of potassium iodide in 200 mL of water and cool.
2. Dissolve 50 g of sodium hydroxide in 250 mL of water and cool.

Add the cold solution 2 to the cold solution 1 and make up to 500 mL. Allow the mixture to stand and decant the clear supernatant (stable for many months) for use. Store in dark brown bottles away from the light.

Method

Add the reagent (3 drops) to the sample (3 drops), agitate and heat the mixture to 100°C in a water-bath, examining it every minute for 10 min. A blank solution should be treated similarly at the same time.

Indications

A brown–orange colour is produced quickly by aliphatic amides and thioamides. The presence of an aromatic ring slows the reaction. The nearer the amide group is to the ring, the more the reaction is inhibited. Substituents in the ring may cause a weak reaction. An immediate black colour is produced by substances that contain *ortho*- or *para*-hydroxy groups and by substances that contain an –NH–NH– or –NH–NH₂ group in an aliphatic side-chain. Some compounds must be heated to 100°C to produce blackening. Colours are given in Table 30.26.

Ninhydrin**Reagent**

Dissolve 0.5 g of ninhydrin in 40 mL of acetone.

Method

Dissolve the sample in methanol, place a drop of the solution on a filter paper, add 1 drop of the reagent and dry in a current of hot air.

Indications

A violet colour that appears rapidly indicates the presence of an aliphatic primary amine or an amino acid group. The presence of an aromatic ring inhibits the response, and the inhibition increases the nearer the amino group is to the ring, as for amphetamine (pink–orange), procainamide and proxymetacaine (both yellow). If the amino group is associated with a saturated ring, a positive but weak pink–violet colour is

Table 30.25 Colours of steroids with naphthol-sulfuric acid

Colour with hot reagent	Steroid	Colour after dilution
Red	Mestranol	Red
Orange-red	Desoxycortone	Blue-black
	Dydrogesterone	–
	Hydroxyprogesterone	Blue, violet (dichroic)
	Noretynodrel	Brown-red
Brown-red	Ethinylestradiol	Pink
Orange	Norethisterone	Orange-brown
Orange, green (dichroic)	Norethandrolone	Red-orange
Yellow	Diethylstilbestrol	Orange
	Testosterone	Green, brown (dichroic)
Green-yellow	Fluoxymesterone	Yellow
Green	Beclometasone	Brown-yellow
	Fluocinolone	Yellow
Yellow-green	Dexamethasone	Yellow
Green, yellow (dichroic)	Estriol	Orange
	Estrone	Orange
	Triamcinolone	Yellow
Green, brown (dichroic)	Fludroxycortide	Yellow
Blue-green, yellow (dichroic)	Estradiol	Orange
Violet	Fludrocortisone	Brown
Brown	Oxymetholone	Pink-orange
	Prednisolone	Brown
	Prednisone	Orange
	Progesterone	Yellow
Red-brown	Dimethisterone	Brown-green
	Enoxolone	Orange
	Fluocortolone	Red-brown
Orange-brown	Alfadolone	Orange
	Androsterone	Orange
	Cortisone	Orange
	Dienestrol	Yellow
Yellow-brown	Carbenoxolone	Orange
	Cholesterol	Violet
	Hydrocortisone	Yellow-brown
Green-brown	Betamethasone	Orange-brown

obtained (amantadine, rimantadine). Gentamicin gives a violet colour after heating for 4 min.

Nitric acid, fuming**Method**

Mix the sample with 3 drops of fuming nitric acid, heat at 50°C for 30 s and observe any colour produced. Cool the mixture, add 2 drops of it to 2 mL of concentrated sulfuric acid and observe the colour. To the remainder of the cooled mixture, add 2 mL of water followed by 2 mol/L sodium hydroxide, dropwise, until pH 8 is reached (use an indicator paper).

Indications

Chlorinated phenols give a series of colours in the three parts of this test (Table 30.27).

Nitric acid-sulfuric acid (Erdmann's reagent)

Mix 1 mL of concentrated nitric acid with 30 mL of concentrated sulfuric acid.

Table 30.26 Colours with Nessler's reagent

Colour	Compound
Orange	Acebutolol (slow), carbidopa (→black), methotrexate
Brown-orange	Acetylcarbromal (slow), bromvaletone, carbromal, chloramphenicol, dinitolmide, etenzamide (weak), ethionamide, fluoroacetamide, nicotinamide, phenacemide (slow), pheneturide (slow), protionamide, pyrazinamide, salicylamide (weak), urea
Yellow	Dihydrostreptomycin (→brown), penicillamine
Brown	Demeclocycline, mebutamate (slow), nadolol, paracetamol (slow)
Yellow-brown	Atenolol (slow)
Black (immediate)	Adrenaline, apomorphine, ascorbic acid, benserazide, dihydralazine, dobutamine, dodecyl gallate, dopamine, ethylnoradrenaline, hexoprenaline, hydralazine, iproniazid, isocarboxazid, isoetarine, isoniazid, isoprenaline, levodopa, mebanazine, methylidopa, methyldopate, nialamide, noradrenaline, phenelzine, procabazine, protokylol, rimiterol
Black (at 100°C)	Cimetidine, gentamicin, labetalol, meprobamate (grey-black), methallibure, salinazid, thiacezone

Table 30.27 Colours with fuming nitric acid

Colour			
Part 1	Part 2	Part 3	Compound
Orange-red	Orange	Orange-brown	Hexachlorophene
Red	Red	Brown-violet	Pentachlorophenol

Method

Dissolve the sample in 1 mL of ethanol, add a pellet of potassium hydroxide and evaporate to dryness (100°C in a water-bath). To the residue add 0.5 mL of water and 1 mL of carbon tetrachloride, shake and allow to separate. Decant the lower carbon tetrachloride layer and shake it with 1 mL of the reagent.

Indications

A red colour in the acid layer suggests the presence of clofenotane or its metabolite, dichlorodiphenyldichloroethylene (DDE). The red colour changes to orange and then to green. Weak pink colours are given by aldrin, dieldrin and endrin. A red colour is also given by dichlorodiphenyldichloroethane (DDD, mitotane), but the colour does not change.

Note that the substance should be tested to ensure that it does not give a colour with sulfuric acid alone.

Nitrous acid**Method**

Dissolve the sample in a minimum volume of water, and add an amount of solid sodium nitrite equal in volume to the sample followed by a few drops of 2 mol/L hydrochloric acid.

Indications

Orange or yellow colours are given by certain sulfonamides, and green, blue or violet colours by certain phenylpyrazolines (Table 30.28).

No response is obtained with succinylsulfathiazole, sulfacetamide, sulfadiazine, sulfadimethoxine, sulfaguanidine, sulfamerazine, sulfaquin-oxaline, sulthiame or propyphenazone.

Palladium chloride**Reagent**

Dissolve, with the aid of heat, 0.1 g of palladium chloride in 5 mL of 2 mol/L hydrochloric acid and dilute the solution to 100 mL with water. Mix together equal volumes of this solution and 2 mol/L sodium hydroxide. The mixed reagent is stable for several weeks.

Table 30.28 Colours with nitrous acid

Colour	Compound
Orange	Sulfafurazole
Yellow	Sulfadoxine, sulfachlorpyridazine, sulfadimidine, sulfaethidole, sulfamethizole, sulfamethoxazole, sulfamethoxydiazine, sulfamethoxypyridazine, sulfametopyrazine, sulfamoxole, sulfaphenazole, sulfapyridine, sulfasomidine, sulfathiazole, sulfinpyrazone
Green	Phenazone
Blue	Dipyron (transient)
Violet	Amidopyrine (transient)

Method

Mix the sample with 1 mL of the reagent and heat at 100°C in a water-bath for 2 min. A blank solution should be treated similarly at the same time.

Indications

Red, orange, yellow, brown or black colours are given by aliphatic compounds that have a sulfur atom in the chain, and by aromatic compounds that have a sulfur atom in the side-chain. However, no colour is given when an S-alkyl chain is present, unless the chain is terminated by an halogenated group.

No response is obtained if the sulfur is in a group that links two rings. Reducing agents such as ascorbic acid, chloral hydrate, chloroform and glucose, and compounds that contain a chain with a hydrazine link (–NH–NH–, –NH–NH₂), give a translucent dark-grey or black colour, but do not give the gradual yellow to orange to brown colour seen with sulfur-containing compounds. Compounds that contain adjacent hydroxyl groups on an aromatic ring give orange colours that turn brown (Table 30.29).

Phosphorus test**Method**

To the sample add 0.5 mL of concentrated nitric acid and 0.2 mL of concentrated sulfuric acid and heat at 100°C in a water-bath for 30 min. Cool, add 1 mL of a 10% (w/v) solution of ammonium molybdate and replace in the water-bath at 100°C for 5 min. A blank solution should be

Table 30.29 Colours with palladium chloride

Colour	Compound
Red	Gloxazone
Orange	Adrenaline (→brown), benserazide (→brown), bitoscanate, captopril, carbidopa (→brown), carbimazole, disulfiram, dobutamine (→brown), ecothiopate, isoetarine (→brown), levodopa (→brown), methallibure, thiamazole, polythiazide, rimiterol, thiacezone, thiopental
Brown-orange	Demeton-S
Yellow	Clindamycin, dazomet, dimercaprol, dimethoate, methisazone (→orange→brown), penicillamine
Orange-yellow	Thialbarbital
Brown	Ambazone, azinphos-methyl, dihydrostreptomycin (slow), ethionamide, malathion, noxythiolin, parathion, phosalone, protionamide, spironolactone, thiram
Orange-brown	Chlorthiamid, diazinon, disulfoton, fenitrothion, formothion, phorate, vamidothion
Black-brown	Di-allate, dichlofluamid, tri-allate
Grey	Chlorfenvinphos
Black	Ascorbic acid, captan, chloroform, cloral hydrate, mebanazine, nifuratel, phenelzine, procabazine, sulfasalazine, sulfaurea, trichlorfon

Table 30.30 Colours with potassium dichromate

Colour	Compound
Red	Carbidopa
Yellow (→brown)	Phenol (2 min)
Green (→brown)	Adrenaline, dopamine, hexoprenaline, isoetharine, isoprenaline, levodopa, methyl dopa, methyl dopate, noradrenaline, rimiterol
Blue-green	Aniline (2 min)
Brown	Benserazide, <i>o</i> -cresol (30 s), <i>m</i> -cresol (2 min), orciprenaline (slow), protokylol (→red-brown on warming), terbutaline (slow)
Green-brown	Dobutamine

treated at the same time. For some compounds, the reaction may occur after shorter heating times than those stated above.

Indications

A bright yellow solution or precipitate indicates the presence of phosphorus and suggests an organophosphorus pesticide, especially if the sample is a water-immiscible liquid. Cyclophosphamide and triclofos also react.

Potassium dichromate

Method 1

Dissolve the sample by shaking in 0.5 mL of 2 mol/L hydrochloric acid and add a few crystals of potassium dichromate.

Indications

An immediate brown colour, or a green colour that changes to brown, indicates the presence of an aminophenol or of a phenol that has two or more hydroxyl groups in adjacent positions on the ring (Table 30.30). Monophenols, halogenated phenols and phenols with hydroxyl groups *meta* to each other react more slowly or not at all.

Method 2

If the sample is a liquid, add 1–2 drops to 1 mL of water followed by 1 mL of a saturated solution of potassium dichromate in 50% v/v sulfuric acid.

Indications

A green colour is given by acetaldehyde, ethanol, methanol, propan-1-ol and propan-2-ol.

Schiff's reagent

Reagent

Dissolve 0.2 g of basic magenta (fuchsin, CI 42510) in 120 mL of hot water, cool, add 20 mL of a 10% (w/v) solution of sodium hydrogen-sulfite and 2 mL of concentrated hydrochloric acid, and dilute to 200 mL. Store at 4°C and protect from light.

Method

Add the sample to 1 mL of the reagent.

Indications

A violet colour indicates the presence of an aliphatic aldehyde. The longer the carbon chain, especially if it is branched, the weaker the response to the test.

Scott's test (see also Cobalt thiocyanate)

Reagent

1. Cobalt thiocyanate dissolved in water (2% w/v) and then diluted 1 : 1 with glycerine
2. Concentrated hydrochloric acid
3. Chloroform.

Method

Add a small amount of the sample to be tested to a test-tube, add 5 drops of solution 1 and shake. If cocaine is present a blue colour develops at

once. Add 1 drop of solution 2 and shake (the blue colour disappears and a clear pink solution develops). Add several drops of solution 3.

Indications

The chloroform layer develops an intense blue colour if cocaine is present. Methadone also reacts.

The LOD is 60 µg cocaine-HCl and 15 µg methadone-HCl.

Simon's test (modified sodium nitroprusside test)

Reagent

1. Dissolve 1 g of sodium nitroprusside in 100 mL of water and add 2 mL of acetaldehyde to the solution with thorough mixing.
2. Freshly prepared 2% sodium carbonate in distilled water.

Method

Add 1 drop of solution 1 to the sample, followed by 2 drops of solution 2.

Indications

A dark-blue colour indicates a secondary amine (e.g. metamfetamine, ephedrine, 3,4-methylenedioxymetamfetamine (MDMA)) or an unsubstituted heterocyclic amine as its free base. A deep blue colour indicates the presence of metamfetamine. Primary amines (e.g. amfetamine, methylenedioxymetamfetamine (MDA)) yield a slow pink to cherry-red colour.

Sodium dithionite

Reagent

A 5% (w/v) solution of sodium dithionite in a 10% (w/v) solution of sodium hydroxide.

Method

Apply the reagent to the sample, either on a white tile or as a solution in a test-tube. A blank solution should be treated similarly at the same time.

Indications

Colours are produced by bis(pyridyl) compounds (Table 30.31). Dark colours are likely to be given by certain metallic solutions because of reduction.

Sodium hypobromite test (for carbamazepine)

Reagent

Dissolve 0.5 mL of bromine in 5 mL of a 40% w/v solution of sodium hydroxide with shaking and cooling. (this should be freshly prepared).

Method

Add 1 mL of 2 mol/L hydrochloric acid to 5 mL of sample and 5 mL of chloroform. Vortex mix for 1 min and centrifuge for 5 min. Remove the upper layer, add 1 mL of the chloroform extract to 0.2 mL of sodium hypobromite reagent and mix for 30 s.

Indications

Carbamazepine forms a blue to violet colour in the chloroform layer. The test has a sensitivity of 250 mg/L.

Sodium nitroprusside

Reagent

A 1% (w/v) solution of sodium nitroprusside.

Method 1

Add the sample to 2 mL of the reagent followed by a drop of 2 mol/L sodium hydroxide.

Indications

Orange colours are given by ketones and red colours by acetaldehyde.

Table 30.31 Colours with sodium dithionate

Colour	Compound
Green	Diquat
Blue	Paraquat

Method 2

Mix the sample with a minimum volume of 2 mol/L sodium hydroxide, evaporate to dryness, dissolve the residue in 2 drops of water and add 0.5 mL of the reagent.

Indications

A violet colour is given by substances that contain labile sulfur in the molecule and by unsubstituted dithiocarbamates.

Method 3

Carry out Method 2 above, but after evaporation to dryness heat the residue until it is yellow or orange in colour before proceeding.

Indications

A violet colour is given by certain substances that contain labile sulfur and do not react to method 2 (e.g. clomethiazole, lincomycin and monosulfiram).

Sodium nitroprusside-acetone**Reagents**

1. Dissolve 2 g of sodium nitroprusside in 5 mL of water and add 45 mL of methanol
2. 2% (w/v) sodium carbonate
3. Acetone
4. 10% acetaldehyde.

Method 1

Add a drop of solution 1 followed by a drop of solution 2 to 3–4 mg of sample dissolved in solution 3 on a spot plate. A purple colour is indicative of amfetamine. The LOD is 30 µg.

Method 2

Add a drop of solution 1 followed by a drop of solution 4 to 1–2 mg of sample dissolved in solution 3. An immediate blue colour is indicative of metamfetamine. The LOD is 5 µg.

Sodium picrate (Steyn test)**Reagent**

Prepare a solution of 5 g sodium bicarbonate and 0.5 g picric acid in 100 mL of water.

Method

Mix the sample with a few drops of chloroform and concentrated sulfuric acid to hasten the reaction while holding a piece of filter paper, impregnated with the reagent, in the vapours that issue from the tube, and heating the contents to 30°C.

Indications

The yellow colour of the filter paper changes from orange to brown–orange and then to orange–red or red in the presence of cyanide. Positive results are given by compounds that contain cyanide groups (e.g. cimetidine, diphenoxylate and isoaminile).

Sulfuric acid**Method**

Apply concentrated sulfuric acid directly to the sample on a white tile or in a test-tube.

Indications

A range of colours is obtained with compounds of various types. Steroids give orange or yellow colours, many of which fluoresce under UV light ($\lambda = 350$ nm) either immediately or after dilution (Table 30.32). Thioxanthenes give red or orange colours that fluoresce under UV light ($\lambda = 350$ nm) (Table 30.33).

Sulfuric acid–fuming sulfuric acid**Reagent**

Mix together 7 mL of concentrated sulfuric acid and 3 mL of fuming sulfuric acid.

Method

Dissolve the sample in a minimum volume of toluene and add 1 or 2 drops of the reagent.

Indications

A red colour that appears in the lower acid layer indicates the presence of dieldrin (colour develops quickly) or aldrin (colour develops slowly). A pink–orange colour is obtained with endrin.

Tetrabromophenolphthalein ethyl ester**Reagent**

Dissolve 50 mg tetrabromophenolphthalein ethyl ester (TBPE) in 100 mL chloroform, shake the solution for 2 min with 1 mL of 10% (v/v) hydrochloric acid and discard the aqueous phase. Dry the organic layer with anhydrous sodium sulfate. Separate the drying agent by filtration. Store the reagent in an amber bottle at 4°C.

Method

Place 0.5 mL of sample to be tested in a conical test tube, add 100 µL phosphate buffer (10 mmol/L, pH 8.0) and vortex mix. Add 50 µL of the TBPE reagent and vortex mix. After 2–3 min note the colour of the chloroform layer. If the sample to be tested is a solid, dissolve 1–2 mg of the material in 0.5 mL of buffer and proceed.

Indications

A deep blue colour indicates quaternary ammonium compounds. An orange, brown, red or purple colour indicates the presence of basic drugs. This test is most sensitive to tertiary amines (e.g. tricyclics, propoxyphene, phenothiazines, diphenhydramine, phencyclidine, methadone, pethidine). Its LOD is 1 mg/L.

Thalleioquin test**Method**

Dissolve the sample in a minimum volume of 2 mol/L hydrochloric acid, add 2 drops of bromine solution, place 1 drop of the mixture on a piece of filter paper and expose the paper to ammonia fumes.

Indications

A green colour indicates the presence of a quinine-type structure (e.g. hydroquinidine, hydroquinine, quinidine, quinine). Cinchonidine and cinchonine do not respond.

Trinder's reagent (see Ferric chloride)**Reagent**

The solution is prepared as follows: 40 g of mercuric chloride and 40 g of ferric nitrate are dissolved in 850 mL of distilled water; 10 mL of concentrated HCl is added and the solution is diluted to 1 L. This solution is stable for 1 year.

Method

A few drops of the reagent are added to a few drops of urine. A purple colour indicates the presence of a salicylate. This test was devised for the quantitative assay of salicylates in serum, with the mercuric chloride serving as a protein precipitant. The ferric chloride test has been modified for use on blood samples (Asselin, Caughlin 1990). Blood (0.5 mL) is mixed with 1.0 mL of acetone and vortex mixed. The sample is centrifuged, and 50 µL of the supernatant is added to 50 µL of ferric chloride. A purple colour at the interface indicates salicylates.

Vanillin reagent**Reagent**

Dissolve 1 g of vanillin in 20 mL of concentrated sulfuric acid, warming if necessary.

Method

Add 2 drops of the reagent to the sample, heat in a water-bath at 100°C for 30 s and note any colour that is produced. Dilute the cooled mixture by adding a few drops of water and note any change of colour.

Indications

Many compounds of different chemical structure react with this reagent. However, for barbiturates, the reaction appears to be a steric

Table 30.32 Reactions of steroids with sulfuric acid

<i>Initial colour</i>	<i>Compound</i>	<i>Fluorescence at 350 nm</i>	<i>Fluorescence after dilution</i>
Red			
Orange-red	Dienestrol	Nil	Nil
	Dimethisterone	Nil	Yellow
	Mestranol	Yellow	Orange (pink in daylight)
Pink			
Orange-pink	Dexamethasone	Nil	Nil
	Prednisolone	Nil	Green (red in daylight)
Orange	Beclometasone (slow)	Nil	Nil
	Cholesterol	Nil	White
	Dydrogesterone	Green-yellow	Green-yellow
	Fludrocortisone	Green	Green (dichroic in daylight)
	Norethandrolone	Green-yellow	-
	Norethisterone	Orange	Orange (violet in daylight)
	Norethynodrel	Orange	Orange
	Oxymetholone	Nil	Nil
	Spironolactone (→yellow-green)	Yellow-green	Green
	Diethylstilbestrol	Nil	Nil
	Triamcinolone	Nil	Nil→green (slow)
Red-orange	Ethinylestradiol	Orange	Orange (red in daylight)
Pink-orange	Betamethasone	Nil	Nil
Green, orange (dichroic)	Hydrocortisone	Green	Green
Yellow	Alfadolone	Nil	Nil
	Androsterone	Nil	White
	Carbenoxolone	Nil	Yellow
	Cortisone	Green	Green
	Desoxycortone	Green-yellow	Yellow (violet in daylight)
	Enoxolone	Nil	Green-yellow
	Fluocinolone	Green	Quenched
	Fluoxymesterone	Green	Quenched
	Fludroxycortide	Green	Quenched
	Hydroxyprogesterone	Green	Quenched
	Estradiol	Green	Green (orange in daylight)
	Prednisone	Green	Green
	Progesterone	Green	Quenched
Orange-yellow	Fluocortolone	(Weak)	(Weak)
Green-yellow	Estrone	Green	Green-yellow (orange in daylight)
No colour	Estriol	Yellow-green	Quenched (orange in daylight)
	Testosterone	Green	Nil

phenomenon that depends on the structure of the side-chain at the 5-position. Dark colours, which are either dispelled or changed to violet, blue or green by dilution, are produced when either side-chain is greater than two carbon atoms in length or contains a cycloalkene ring. Branching can be proximal to the pyrimidine ring, but not distal. No colour is obtained if both side-chains are less than three carbon atoms in length or if either is branched distally or contains an aryl nucleus. Long, straight, saturated chains also appear to hinder reaction.

Hydroxybarbiturates give positive responses (Table 30.34), but bemegride, glutethimide, phenytoin and primidone do not respond. No response is obtained with amobarbital, aprobarbital, barbital, butobarbital, enallylpropymal, hexethal, ibomal, idobutal, metharbital, methylphenobarbital, nealbarbital, phenobarbital or phenylmethylbarbituric acid. With cold reagent, an orange colour is produced by pentobarbital, secobarbital and thiopental, and a brown colour by cyclopentobarbital.

Zwicker reagent (alkaline cobalt test)

This is a general test for barbiturate-like compounds.

Reagent

1. Dissolve 0.5 g of copper(II) sulfate pentahydrate in 100 mL of distilled water.
2. Add 0.5 mL of pyridine to 95 mL of chloroform.

Method

Add a few drops of solution 1 to the sample to be tested, followed by a few drops of solution 2 and then heat.

Indications

The presence of a violet-blue colour indicates barbiturates (Table 30.35). The LOD is 1000 µg for phenobarbital.

Table 30.33 Colours of thioxanthenes with sulfuric acid

Colour	Compound ^(a)
Purple	Oxytetracycline, tetracycline
Red	Caramiphen (when warmed), dantron, fenitrothion, methenamine (when warmed), mequitazine (slowly at 100°C), methacycline, methylprednisolone (after 1 min), metopimazine, nuarimol, ouabin, phenothiazines, prednisolone (after 1 min), pipoxolan
Orange-red	Oxprenolol*, quinomethionate
Violet-red	Morantel, oxytetracycline
Pink	Doxylamine, indapamide (slow)
Orange	Alprenolol, amitriptyline, benactyzine, benzilium, benzquinamide, benzyl nicotinate, chlorprothixine*, clidinium, clopenthixol*, cyclothiazide (→red-brown), diethylthiambutene, diphepanil, diphenhydramine, diphenidol, doxepin, flupentixol*, indometacin, mazindol, mebanazine, mecoprop, methapyrilene, methindizate, methixene*, methyclothiazide, methylpiperidyl benzilate, naproxen, nefopam, nifedipine, nortriptyline, orphenadrine, penthienate methobromide, polythiazide, pyrantel (→violet), rotenone, tiotixene*, tofenacin
Yellow	Acebutolol, amiloride, amiodarone, benzthiazide, broxaldine, broxyquinoline, cinchophen, clefamide, clemastine (green rim), cyclopenthiazide, diphenadione, doxycycline, enoxolone, fenbufen, furosemide, halquinol, hydroquinidine*, hydroquinine*, lorazepam, methyl benzoquate, 5-methyltryptamine, α -methyltryptamine, <i>N</i> -methyltryptamine, metolazone, minocycline, piperacetazine (→red), procyclidine, quinidine*, quinine*, rimiterol, salbutamol, salinazid, sodium cromoglicate, trichlormethiazide, veratrine (→violet), zomepirac
Orange-yellow	Ethyl biscoumacetate, hexoprenaline, pizotifen (→violet)
Green	Bromodiphenhydramine, cyclizine, diphenhydramine, diphenylpyraline, phenothiazine, protriptyline
Blue	
Brown-blue	Chlortetracycline, demeclocycline, thioridazine
Violet	Bendroflumethiazide, chlorotrianisene, chromonar*, clofazimine, cyproheptadine, dosulepin, mesoridazine (→blue), methylenedioxymfetamine, nicergoline, perazine, phenindione, rolitetracycline, tetracycline
Red-violet	Trifluomeprazine
Brown	Chelidonine, sulindac
Red-brown	Lymecycline
Orange-brown	Biperiden, ouabain (slow)
Yellow-brown	Tylosin (slow)
Grey-brown	Octaphonium
Black	
Blue-black	Clomocycline

^(a)Compounds giving colours that fluoresce under UV light ($\lambda = 350$ nm) are indicated by an asterisk.

Basic tests for drug substances and products

The WHO has published texts *Basic Tests for Drugs* (WHO 1998), which includes pharmaceutical substances, medicinal plant materials and dosage forms, and *Basic Tests for Pharmaceutical Substances* (WHO 1986). The basic tests described, which are designed to verify the identity of drug substances and medicinal products and to detect gross contamination, use a limited number of readily available reagents and equipment. Overall, the combined texts offer compound-specific tests for approximately 500–600 products that are based on a combination of organoleptic checks and simple physicochemical tests, such as colour reactions and melting-point determinations.

Table 30.34 Reactions of barbiturates with vanillin reagent

Colour after heating	Compound	Colour after dilution
Red	3'-Hydroxybutobarbital	Violet (transient)
Violet-red	Heptobarbital	Colourless
	3'-Hydroxyamylbarbital	Colourless
Brown-red	Cyclobarbital	Green
	Cyclopentobarbital	Green
	Pentobarbital	Violet
	Secobarbital	Violet
	Thiopental	Violet
Orange	Butalbital (weak)	Colourless
	Secbutobarbital (weak)	Violet
Brown-orange	Allobarbital	Violet (transient)
	Brallobarbital	Brown-orange
	Talbutal	Violet
	Thialbarbital	Violet (transient)
Brown	Hexobarbital	Violet
Violet-brown	Methohexital	Colourless
	Vinbarbital	Colourless
	3'-Hydroxypentobarbitone	Colourless

Table 30.35 Colours with Zwicker reagent

Colour	Compound
Blue	Diphenylhydantoin
Green	Diacetylmorphine, pseudoephedrine
Yellow	Tetracycline

Semi-quantitative TLC methods have been developed as basic tests using a limited number of solvent systems and detection systems. References to these tests are given in (WHO 1998).

It should be remembered that basic tests are not, in any circumstances, intended to replace pharmacopoeial requirements, but should be used as a rapid, inexpensive means to verify identity and strength of drugs and medicinal products, and possibly to detect poor-quality counterfeit and other substandard products. In the event that suspect products are detected, these should be tested for compliance against pharmacopoeial requirements.

Colour reagents and thin-layer chromatography

Many common colour-test reagents are used routinely as spray locating reagents in TLC (e.g. acidified iodoplatinate solution, Dragendorff's reagent, Marquis reagent, Van Urk's reagent). It is worth noting that the preparation of the spray equivalent of a colour reagent may differ slightly from that of the colour reagent preparation itself (see Chapters 1 and 11 and Index of Reagents).

It is also true that the reaction and resultant colour that marks the presence of a certain substance obtained from spraying a reagent on TLC plates may differ from that obtained from a direct colour test and, in some cases, will not yield any results. This is because of pH effects – that is, whether the TLC plate has been dipped in 0.1 mol/L sodium hydroxide and the substance tested is acidic in character (i.e. the free acid of a salt post extraction). However, this aspect of colour reagents and their use in TLC as sprays can often give clues as to the drug or substance's chemistry.

A list of substances and their colour reactions to various TLC spray reagents is given in the Index of Analytical Data.

Appendix 30.1. Colour reactions of amphetamine-like compounds

<i>Compound</i>	<i>Marquis</i>	<i>Mecke</i>	<i>Froehde</i>	<i>Mandelin</i>
D-Amphetamine	R/O→R/Br	Y	Css	O→R→G/Br
Benzphetamine	R→Br	G→Y	Y→G	
Diethyltryptamine	Y→Br	B→G/Bk		
2,5-Dimethoxyamphetamine	IrY→R/Br	Br/G	ItG	ItG
Dimethoxymethylamphetamine	Y/G	Y→G→dkR/Br	Y/G	Y→G→Br
2,3-Dimethylamphetamine	M R/V	V dkV	V Gy/V	V Gy/G
2,4-Dimethylamphetamine	V O/R	G G/Br	Y Br/Y	Y Br/Y
2,5-Dimethylamphetamine	Y/G	Br/G	Y/G	Y/G
2,6-Dimethylamphetamine	Css	Br/G	Css	Br/Y
3,4-Dimethylamphetamine	Y/Br	G→Br/G	Lt G	NR
3,5-Dimethylamphetamine	Y	O/Br→Br/G	Css	Gy/V
Dimethyltryptamine	Y→G→Br	Y→G/Bk	ItY	G→Br
Ephedrine	Css	Css	Css	ItBr
Mephentermine	O→RBr	G		
Mescaline	O/R	O→G/Br	ItY	Y→G→Br
Metamphetamine	R/O→R/Br	NR	NR	G→B/G
3-Methoxy-4,5-methylenedioxyamphetamine	O→M	dkB	B→dkB	O→O/R
3,4-Methylenedioxyamphetamine	B/BK→dkV	G→dkB	G→dkV	R/V→dkV
3,4-Methylenedioxymethylamphetamine	B→V→Bk	G→dkB	Y/G→dkB	B→V→Bk
Paramethoxyamphetamine	Effervescence	ItG	G→R/Br	
Phendiamine	V→Bk	ItG	ItV	G
Phendimetrazine	NR	NR	NR	NR
Pheniramine	NR	NR	NR	NR
Phenmetrazine	NR	ItY	NR	NR
Phentermine	ItO	ItO	Br	G
Phenylephrine	O	O/Br	B	G
Phenylpropanolamine	ItY	ItY	Br	NR
Pseudoephedrine	Css	Css	Css	ItBr
3,4,5-Trimethoxyamphetamine	R/O→O	G/Br→Br	Y	NR

B, blue; Bk, black; Br, brown; Css, colourless; dk, dark; G, green; Gy, grey; It, light; M, magenta; NR, no reaction; O, orange; R, red; V, violet; Y, yellow.

Appendix 30.2. Colour reactions of narcotics

<i>Compound</i>	<i>Marquis</i>	<i>Mecke</i>	<i>Froehde</i>	<i>Mandelin</i>
Anileridine	O	NR	NR	NR
Codeine	V	dkG	ItG	ItG
Dihydromorphine	V/Br	Br	V	Css
Ethylmorphine	O→V	dkG	Y→ItG→B	G
Fentanyl	O→Br	G		
Diamorphine	V	ItG	M	NR
Hydrocodone	V	G→B	ItY	NR
Hydromorphine	Y→R→V	Y→O→G	dkB→Gy	
Levorphanol	Gy→Bk	ItGy→Bk	B	G
Meperidine (pethidine)	O(slow)	ItY(slow)	NR	NR
Methadone	ItV	G→Br	B	
Morphine	M→V	G→B	V→Gy	ItGy
Nalorphine	O→R→V	ItG→dkG	M	G
Oxycodone	O/Y→ItV	Y→dkG	Y→YBr	Y→G
Oxymorphone	M→V	Y→Br	B→V	Bk
Papaverine	M	dkG	G	G/Br
Pentazocine	M	ItGy→V	dkB	G→Br
Thebaine	V/Br	Br	Br	V/Br

B, blue; Bk, black; Br, brown; Css, colourless; dk, dark; G, green; Gy, grey; It, light; M, magenta; NR, no reaction; O, orange; R, red; V, violet; Y, yellow.

Appendix 30.3. Chemical test for gamma-hydroxybutyric acid (GHB) and gamma-butyrolactone (GBL)

Reagents

Chlorophenol red

0.04 g chlorophenol red in 100 mL water, adjust to pH 7 with 0.01 mol/L sodium hydroxide.

Modified Schweppes

1. 2 g of dextrose in 20 mL of water
2. 2.4 g of aniline hydrochloride in 20 mL of ethanol.

Mix solutions 1 and 2 together and dilute to 80 mL with methanol.

Bromocresol purple

0.04 g bromocresol purple in 100 mL of water. Adjust to pH 7 with 0.01 mol/L sodium hydroxide.

Bromothymol blue

0.04 g bromothymol blue in 100 mL of water. Adjust to pH 7 with 0.01 mol/L sodium hydroxide.

Bromocresol green

0.03 g bromocresol green in 100 mL of methanol–water 4 : 1. Adjust to pH 7 with sodium hydroxide 0.01 mol/L.

Methyl orange

0.01 g methyl orange in 100 mL of methanol. Adjust to pH 7 with sodium hydroxide 0.01 mol/L.

Colour test 1

Reagent

Mix chlorophenol red and modified Schweppes reagent in a 3 : 1 ratio.

Method

Place 0.5 mL of a liquid sample in a test tube. Adjust to pH 5–8 with 0.01 mol/L sodium hydroxide. Add 2 drops of the test reagent and swirl. An immediate colour change (orange–red to dark red) indicates GHB. GBL gives a yellow colour.

Colour test 2

Reagent

Mix bromocresol purple and bromothymol blue in a 1 : 1 ratio and mix the combined reagent with modified Schweppes reagent in a 7 : 1 ratio.

Method

Same procedure as for colour test 1. GHB gives a purple colour and GBL gives a yellow colour.

Colour test 3

Reagent

Mix bromocresol green and methyl orange in a 1 : 1 ratio and mix the combined reagent with modified Schweppes reagent in a 3 : 1 ratio.

Method

Adjust the pH of the test solution to neutral if necessary and add 2 drops of test reagent. A dark-green colour indicates GHB; GBL gives a yellow–orange colour.

Colour test 4

Reagent

1% cobalt nitrate solution.

Method

Place 0.5 mL of a liquid sample in a test tube. Add a few drops of the test reagent. A pink-to-violet colour indicates GHB.

Appendix 30.4. Tests for metals and anions not covered by the general scheme

Metals

Reinsch test (for antimony, arsenic, bismuth and mercury)

Method

Use a 5 × 5 mm piece of copper foil or mesh. Clean the copper in 50% (v/v) nitric acid until it develops a shiny surface, rinse with water and proceed as follows:

1. Place 20 mL of sample and 10 mL of concentrated hydrochloric acid in a 100 mL conical flask and add the copper foil or mesh.
2. Heat on a boiling water-bath for 1 h and add as necessary dilute hydrochloric acid to maintain the volume of the solution.
3. Cool, remove the copper, wash gently with water and examine the surface.

Indications

- A purple–black stain indicates antimony.
- A dull black stain indicates arsenic.
- A shiny black stain indicates bismuth.
- A silvery deposit suggests mercury.

Note that other elements, (e.g. selenium and tellurium) also give black deposits and sulfur compounds may give a speckled discoloration.

Confirmatory test

This is applied to the stained copper foil or mesh derived as described above.

Reagents

1. Aqueous potassium cyanide solution (100 g/L)
2. Freshly prepared aqueous sodium sulfite solution (50 g/L)
3. 3 mol/L nitric acid
4. Quinine/potassium iodide reagent. Dissolve 1 g of quinine in 100 mL of water containing 0.5 mL of concentrated nitric acid. When the quinine has dissolved add 2 g of potassium iodide.

Method

1. Leave the copper in potassium cyanide solution for 10 min.
2. Wash any undissolved stain with water and add 1 mL of sodium sulfite solution followed by 1 mL of 3 mol/L nitric acid.
3. Shake the mixture frequently for 5 min and add 1 mL of water and 1 mL of potassium iodide reagent.

Indications Stains caused by the presence of arsenic dissolve in potassium cyanide solution whereas stains due to antimony and bismuth remain. The slow formation of an orange–brown suspension is seen with the quinine/potassium iodide reagent if a stain due to bismuth is present. The LOD for arsenic is about 5 mg/L and for antimony and bismuth is 2 mg/L. A more specific test for arsenic, the ‘Gutzeit test’, involves the conversion of arsenic to arsine and subsequent reaction of the gas with reagents such as silver diethyldithiocarbamate to give a coloured product. A modified version of the Gutzeit apparatus is available from Fischer Scientific which allows quantitative measurement of arsenic in stomach contents, food, water and other materials.

Confirmatory test for mercury

This is applied to the silver stained foil or mesh from the Reinsch test.

Reagent Cuprous iodide suspension. Dissolve 5 g of copper sulfate and 3 g of ferrous sulfate in 10 mL of water with continuous stirring and add 7 g of potassium iodide dissolved in 50 mL of water. Allow the cuprous iodide precipitate to form, filter, and wash with water. Transfer the precipitate as a suspension in water to a brown glass bottle.

Method Add 0.1 mL of copper(I) iodide suspension to a filter paper, place the foil on the suspension, cover and leave for 1–12 h.

Indications A salmon-pink colour due to the formation of cuprous mercuric iodide suggests the presence of mercury and positive results may appear within 1 h, but with low concentrations colour development may take up to 12 h. The LOD for mercury is about 5 mg/L.

Barium**Reagents**

1. Concentrated hydrochloric acid
2. Platinum wire.

Method

1. Dip the end of the platinum wire into the hydrochloric acid and then into the test material.
2. Insert the wire into the hot area of a micro-burner flame and observe any changes in the flame colours.

Indications

Barium salts impart a green flame; copper and thallium salts also give a green flame in this test.

Confirmatory test**Reagents**

1. 1 mol/L sulfuric acid
2. 100 g/L aqueous lead acetate solution
3. 50 mL/L aqueous acetic acid
4. Solid ammonium acetate.

Method

1. To a mixture of 2 mL of lead acetate solution and 2 mL of dilute sulfuric acid add enough ammonium acetate to dissolve the lead acetate precipitate.
2. Add 0.1 mL of dilute acetic acid to 1 mL of sample followed by 1 mL of the lead sulfo-acetate solution (from step 1) and vortex mix for 5 s.
3. Centrifuge for 2 min and observe the tube against a dark background.

Indications Barium salts yield either a white turbidity or white precipitate. Calcium and strontium salts interfere. The LOD for barium is approximately 100 mg/L.

Copper**Reagents**

1. 10 g/L solution of dithiooxamide in methanol
2. Concentrated ammonium hydroxide.

Method

Place 0.1 mL of sample onto a filter paper to give a spot of around 1 cm in diameter. Expose the paper to ammonia fumes and add 0.1 mL of the dithiooxamide reagent to the spot.

Indications

An olive green stain is seen with copper salts. A green stain due to chromium is usually visible before addition of the dithiooximide reagent. Yellow-brown or yellow-red colours are given by several other metals. The LOD for copper is approximately 1 mg/L.

Iron**Reagents**

1. 2 mol/L hydrochloric acid
2. 10 g/L aqueous potassium ferricyanide solution
3. 10 g/L aqueous potassium ferrocyanide solution.

Method

1. Add 0.1 mL of 2 mol/L hydrochloric acid and 0.05 mL of potassium ferricyanide solution to 0.1 mL of sample.
2. Add 0.1 mL of 2 mol/L hydrochloric acid and 0.05 mL of potassium ferrocyanide solution to a further 0.1 mL of sample.
3. Agitate both mixtures for 5 s, leave for 5 min at room temperature and centrifuge for 5 min.

Indications

Ferrous salts give a deep blue precipitate with potassium ferricyanide and ferric salts give a deep blue precipitate with potassium ferrocyanide. The LOD for both ferrous and ferric salts is about 10 mg/L.

Thallium**Test 1****Reagents**

1. Cyanide reagent: dissolve 1.6 g of sodium hydroxide, 1.2 g of potassium sodium tartrate and 1.36 g of potassium cyanide in 10 mL of water.
2. Prepare a fresh solution of dithizone (250 mg/L) in chloroform.

Method Add 1 mL of cyanide reagent to 5 mL of urine in a stoppered glass test-tube and vortex mix for 20 s. Then add 2 mL of dithizone solution, vortex mix for 1 min and centrifuge (5 min).

Indications The presence of thallium is indicated by a pink-red colour in the chloroform layer. The test will detect thallium at 0.1 mg/L. A number of other metal ions give colours with this test.

Test 2**Reagents**

1. Bromine water (saturated)
2. 6 mol/L hydrochloric acid
3. 20% (w/v) aqueous sulfosalicylic acid solution
4. 0.1% (w/v) aqueous methyl violet solution
5. Toluene.

Method To 1 mL of urine carefully add 2 drops of hydrochloric acid and 5 drops of bromine water. Leave to stand for 5 min, add 5 drops of sulfosalicylic acid solution and 0.5 mL of toluene, and shake gently.

Indications A transient blue-green colour in the toluene layer suggests the presence of thallium. The LOD for thallium is 1 mg/L.

Anions**Borate****Reagents**

1. 10 g/L solution of turmeric in methanol
2. 1 mol/L hydrochloric acid
3. 4 mol/L ammonium hydroxide.

Method

1. Prepare turmeric test papers by soaking strips of filter paper (1 × 5 cm) in the turmeric solution and drying at room temperature.
2. Acidify a portion of the sample with the dilute hydrochloric acid and apply to a strip of turmeric paper.
3. When the paper is dry moisten it with dilute ammonium hydroxide.

Indications

A positive sample will impart a red-brown colour to the turmeric paper which strengthens as the paper dries. When the paper is moistened with dilute ammonium hydroxide a green-black colour is produced. Oxidising agents that bleach turmeric (e.g. bromates, chlorates, iodates and nitrites) interfere with the test. The LOD for borate is 50 mg/L.

Confirmatory test

Reagent 0.5 g/L solution of carminic acid in concentrated sulfuric acid.

Method Add 0.5 mL of filtered stomach contents or scene residue to a 10 mL glass tube and slowly pour 0.5 mL of carminic acid solution down the inside of the tube so that a layer is formed underneath the sample.

Indications Borate is indicated by the formation of a blue-violet ring at the junction of the two layers. Note that strong oxidising agents (e.g. bromates, chlorates, iodates and nitrites) also give positive results.

Bromides**Reagents**

1. 2 mol/L nitric acid
2. 10 g/L solution of silver nitrate
3. Concentrated ammonium hydroxide.

Method

1. To 1 mL of clear tests sample add 0.5 mL of 2 mol/L nitric acid and mix for 5 s.

2. Add 0.1 mL of silver nitrate solution.
3. Centrifuge any precipitate and decant off the supernatant.
4. Add 0.1 mL of concentrated ammonium hydroxide.

Indications

Bromide gives an off-white precipitate that is slightly soluble in ammonium hydroxide. A white precipitate that dissolves in ammonium hydroxide indicates chloride and a yellowish precipitate insoluble in ammonium hydroxide suggests iodide. The LOD for bromide is 50 mg/L.

Confirmatory test**Reagents**

1. Saturated solution of fluorescein in aqueous acetic acid (600 mL/L)
2. Concentrated sulfuric acid
3. Solid potassium permanganate.

Method

1. Soak a strip of filter paper in fluorescein solution.
2. Transfer 2 mL of test solution to a 10 mL glass tube and add about 2 mg of potassium permanganate.
3. Add 0.2 mL of concentrated sulfuric acid and hold the fluorescein paper at the mouth of the tube.

Indications Any bromide is oxidised to free bromine, which then reacts with the fluorescein dye (yellow) to give tetrabromofluorescein (eosin) which has a pink-red colour. The LOD for bromide is 50 mg/L.

Chlorate

Chlorates and other oxidising agents can be detected by the diphenylamine test described previously. The following tests can also be applied.

Test 1**Reagents**

1. 6 mol/L sulfuric acid
2. 1% w/v aqueous solution of indigo carmine
3. Solid sodium sulfite.

Method To 1 mL of sample add 4 mL of 6 mol/L sulfuric acid followed by 1 mL of indigo carmine reagent.

Indications A deep blue colour indicates the presence of chlorates. The colour fades after adding several crystals of sodium sulfite. The same reactions are given by bromates and hypochlorites.

Test 2**Reagents**

1. Manganous sulfate reagent: a 1:1 mixture of saturated aqueous manganous sulfate and orthophosphoric acid
2. A 10 g/L solution of diphenylcarbazide in methanol.

Method

1. Add 0.2 mL of manganous sulfate reagent to 0.1 mL of test sample and warm the mixture gently over a burner.
2. Cool and add 0.1 mL of diphenylcarbazide solution.

Indications Chlorate yields a purple colour that intensifies after cooling and adding diphenylcarbazide. A similar reaction is given by persulfates and periodates.

Note: after ingestion of chlorate the blood turns brown owing to the formation of methaemoglobin by oxidation of the ferrous ion of haemoglobin. Add 0.2 mL of a 10% (w/v) of aqueous potassium cyanide solution to 1 mL of blood and an immediate red colour confirms the presence of methaemoglobin. Exposure to a wide range of other substances (e.g. aniline, nitrites, nitrates, aniline, dapsone, benzocaine, urea herbicides) causes methaemoglobinaemia.

Fluoride**Test 1****Reagents**

1. A 50 g/L aqueous solution of sodium chloride
2. Concentrated sulfuric acid

3. Calcium chloride (solid)
4. Powdered silica.

Method

1. To 5 mL of sample in a porcelain crucible add 100 mg of calcium chloride and gently evaporate to dryness over a burner.
2. Destroy the organic material by strong heating to leave a white ash.
3. Mix the residue with 200 mg of powdered silica.
4. Apply a drop of sodium chloride solution to a microscope slide, add 1 mL of concentrated sulfuric acid to the contents of the crucible and immediately position the slide such that the sodium chloride drop is suspended over the crucible.
5. Rest a small beaker of ice on the slide and heat the crucible gently for 5 min over a burner.
6. Remove the slide after 5 min and examine the sodium chloride solution under a microscope.

Indications In the presence of fluorine-containing compounds, silicon tetrachloride is produced, which dissolves in the sodium chloride solution to form sodium silicon tetrafluoride. This forms small hexagonal crystals as the water evaporates from the slide, which sometimes have a pink hue. The crystals are seen at the edge of the drop and appear before the larger cubic crystals of sodium chloride.

Test 2**Reagents**

1. Concentrated sulfuric acid
2. Calcium hydroxide (solid)
3. Paraffin wax.

Method

1. Repeat steps (1) and (2) described for test 1.
2. Smear a film of paraffin wax on one side of a glass microscope slide and expose part of the surface by making an identifiable sign on the paraffin film.
3. Add 5 mL of concentrated sulfuric acid to the crucible and cover it with the slide, with the paraffin film on the inside.
4. Heat the crucible gently for 20 min and then remove the slide.
5. Remove the paraffin film with toluene and examine the slide.

Indications Hydrogen fluoride is generated from fluorine-containing compounds and etches the glass to give a mark corresponding to that made in the paraffin film.

Both tests have a LOD of approximately 100 mg/L of fluoride.

Hypochlorite**Test 1****Reagents**

1. Glacial acetic acid
2. 50 g/L aqueous lead acetate solution.

Method

1. To 1 mL of test solution add acetic acid dropwise to reach a pH of approximately 6 (test with universal indicator paper).
2. Add 0.5 mL of lead acetate solution and boil for 3 min.

Indications Hypochlorite forms a brown precipitate. An immediate brown to black precipitate is given by sulfides with lead acetate.

Test 2**Reagents**

1. 100 g/L aqueous potassium iodide solution
2. Glacial acetic acid
3. Solid starch.

Method

1. To 0.1 mL of test solution add 0.1 mL of acetic acid followed by 0.1 mL of potassium iodide solution.
2. Mix and add about 20 mg of starch.

Indications Hypochlorite gives a blue colour. (*Note:* hypochlorite also gives a positive reaction in the diphenylamine test for oxidising agents described above.)

Iodides

Reagents

1. 10 g/L aqueous silver nitrate solution
2. 2 mol/L nitric acid
3. Concentrated ammonium hydroxide.

Method

1. To 1 mL of clear test solution add 0.1 mL of nitric acid and 0.1 mL of silver nitrate solution.
2. Centrifuge down any precipitate, remove the supernatant and add to it 0.1 mL of ammonium hydroxide.

Indications

A curdy yellow precipitate that is insoluble in ammonium hydroxide is given by iodides. Chlorides give a white precipitate that dissolves in ammonium hydroxide, and an off-white precipitate that is sparingly soluble in ammonium hydroxide suggests bromides.

Confirmatory test

Reagents

1. 2 mol/L hydrochloric acid
2. 100 g/L freshly prepared sodium nitrite solution
3. Solid starch.

Method Mix thoroughly about 20 mg of starch with 0.1 mL of test solution, 0.1 mL of hydrochloric acid and 0.1 mL of sodium nitrite solution in a test-tube.

Indications A blue colour confirms the presence of iodide.

Oxalates

Reagents

1. 100 g/L aqueous calcium chloride solution
2. 30% (v/v) solution of acetic acid
3. 2 mol/L hydrochloric acid.

Method

1. Add 1 mL of calcium chloride solution to 2 mL of test solution and mix.
2. If a precipitate forms, add 1 mL of acetic acid.
3. If the precipitate does not dissolve, separate it by centrifugation and add 1 mL of dilute hydrochloric acid.

Indications

A white precipitate that is insoluble in acetic acid but dissolves in hydrochloric acid indicates the presence of oxalates.

Confirmatory test

Reagents

1. Thiobarbituric acid
2. Concentrated ammonium hydroxide.

Method

1. Add 50 µL of test solution to 100 µL of ammonium hydroxide in a micro test-tube and mix thoroughly.
2. Gently evaporate the mixture to dryness over a micro-burner.
3. Add about 200 mg of thiobarbituric acid and reheat gently to about 150°C.

Indications Oxalates give a bright red product that is soluble in methanol. The LOD for oxalate is 250 mg/L.

Sulfides

Reagents

1. 10% (v/v) aqueous solution of sulfuric acid
2. 10% (w/v) solution of lead acetate in boiled and purified water
3. 2 mol/L acetic acid.

Method

1. Immerse a strip of white filter paper in a mixture of 10 volumes of lead acetate solution and 1 volume of acetic acid and allow to dry.

2. To 1 mL of test sample in a test-tube add 3 mL of sulfuric acid and insert the lead acetate paper into the neck of the tube.
3. Heat the tube in a boiling water-bath for 5–10 min.

Indications

Hydrogen sulfide fumes that turn lead acetate paper black are produced by sulfides. The LOD for sulfide is 50 mg/L.

Thiocyanates

Reagent

A 50 g/L aqueous solution of ferric chloride.

Method

Mix 0.5 mL of ferric chloride solution with 0.5 mL of sample.

Indications

A deep red colour is given by thiocyanates. The LOD for thiocyanate is 50 mg/L.

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